STACKCALC LEARNING LAB // OFFICIAL TEACHER & PARENT COACHING KEY

Big Ideas Math Middle School: Coach Guide & 3-Visit Spiral Solutions

Complete 10-Chapter, 45-Day Facilitation Playbook, 30-Second Zero-Fail Starters, 5 Variety Game Modes, and Verified SC-6 RPN Stack Snapshots (Grades 6–7 Bridge).

1. 30-SEC ZERO-FAIL + 3-SPEED RAMP	2. 2-NUMBER SNAPSHOT = YOUR WORK	3. 3-VISIT SPIRAL & SKIP SHIELDS
Every day starts with a 30-Second Zero-Fail Starter (+5 XP) you can't get wrong, followed by Problem 1 (60-Sec Quick Win), Problem 2 (Rotating Variety Game), and Problem 3 (Secret-Genius Boss)!	Never copy 6 lines of algebra steps! Your SC-6 4-line elevator holds intermediate numbers on Floor 2 (Y) and Floor 1 (X). Writing down the 2 numbers on Y and X before your final keypress IS your work!	Problems 1 & 2 require ONLY 1 way. On Problem 3, prove 2 different ways match to earn a [2x-STAR]! Every 3 [2x-STAR]s unlocks a [SKIP-SHIELD] (skip any 1 future problem!), and trying all 3 ways unlocks [DEALERS-CHOICE]!

Chapter & Big Ideas Topic	Daily 15-Minute Missions	3-Way Passport	Twin-Way Stars & Rewards
Ch.1: Numerical Expressions & Factors	[ ]D1 [ ]D2 [ ]D3 [ ]D4 [ ]D5	[ ]Build [ ]Pat [ ]Stack	[ ] [ ] [ ]** -> [ ]Shield | [ ]1. Prime Architect
Ch.2: Fractions & Decimals	[ ]D6 [ ]D7 [ ]D8 [ ]D9 [ ]D10	[ ]Build [ ]Pat [ ]Stack	[ ] [ ] [ ]** -> [ ]Shield | [ ]2. Tower Master
Ch.3: Ratios, Rates & Percents	[ ]D11 [ ]D12 [ ]D13 [ ]D14 [ ]D15	[ ]Build [ ]Pat [ ]Stack	[ ] [ ] [ ]** -> [ ]Shield | [ ]3. Ratio Navigator
Ch.4: Algebraic Expressions & Properties	[ ]D16 [ ]D17 [ ]D18 [ ]D19	[ ]Build [ ]Pat [ ]Stack	[ ] [ ] [ ]** -> [ ]Shield | [ ]4. Tile Alchemist
Ch.5: Areas of Polygons & Elapsed Time	[ ]D20 [ ]D21 [ ]D22 [ ]D23	[ ]Build [ ]Pat [ ]Stack	[ ] [ ] [ ]** -> [ ]Shield | [ ]5. Polygon Engineer
Ch.6: Integers, Rational Numbers & The Coordinate Plane	[ ]D24 [ ]D25 [ ]D26 [ ]D27 [ ]D28	[ ]Build [ ]Pat [ ]Stack	[ ] [ ] [ ]** -> [ ]Shield | [ ]6. Zero-Pair Cmdr
Ch.7: Equations & Inequalities	[ ]D29 [ ]D30 [ ]D31 [ ]D32 [ ]D33	[ ]Build [ ]Pat [ ]Stack	[ ] [ ] [ ]** -> [ ]Shield | [ ]7. Balance Breaker
Ch.8: Surface Area & Volume of 3D Solids	[ ]D34 [ ]D35 [ ]D36 [ ]D37	[ ]Build [ ]Pat [ ]Stack	[ ] [ ] [ ]** -> [ ]Shield | [ ]8. 3D Solid Builder
Ch.9: Statistical Measures & Data Displays	[ ]D38 [ ]D39 [ ]D40 [ ]D41	[ ]Build [ ]Pat [ ]Stack	[ ] [ ] [ ]** -> [ ]Shield | [ ]9. Data Scientist
Ch.10: Grade 7 Bridge — Signed Rationals, 2-Step Equations & Probability	[ ]D42 [ ]D43 [ ]D44 [ ]D45	[ ]Build [ ]Pat [ ]Stack	[ ] [ ] [ ]** -> [ ]Shield | [ ]10. Stack Champion

CHAPTER 1: Numerical Expressions & Factors (Days 1–5)

Aligned to Big Ideas Math Grade 6, Chapter 1: Whole Number Operations, Powers & Exponents, Order of Operations, Prime Factorization, GCF & LCM.

Day 1 (Lesson 1.1): Whole Number Division & Leftovers That Actually Make Sense

SPIRAL PATH: Thread A (Factor & Split) • [[BUILDER] VISIT 1: BUILDER WAY — Launch of Thread A] // VARIETY MODE: [SPARK] Mode C: Human vs. Screen Prediction

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Grab 7 Counting Tiles and make groups of 2. You see 3 pairs and 1 leftover! Now press **7 [ENTER] 2 [g] [÷R]** on your SC-6—watch **X: 3** and **Y: 1** pop up together! (**[ ] +5 Spark XP!**)

The Real Question: When you divide things up in real life, what happens to the leftovers—and how can you prove your division is right in 3 button clicks without doing long division twice?

Why Should I Care? (2-Min Hook): Imagine you have 135 skateboard bearings and each board needs 8 bearings (2 per wheel). If you type 135 ÷ 8 into a phone calculator, it spits out **16.875**. What on earth is **0.875** of a steel bearing?! You can't saw a bearing into **.875** pieces! You want to know two real numbers: How many full skateboards can I build? and How many bearings are sitting left over in the tray?
Why Try a New Way Today? (When the Old Way Gets Annoying): Phone calculators spit out ugly decimals like **16.875** that hide the real leftover items. Building tile rows (**[BUILDER]**) shows what a remainder actually is, and **[g] [÷R]** (**[STACK-HACK]**) handles huge numbers in 1 click!

Camera Angle	What You Do	What You See				
Angle 1: Physical Blocks (**COUNT-01**)	Take 23 Counting Tiles and snap them into rows of 4 tiles each.	You get 5 full rows of 4 (= 20), and 3 loose tiles sitting at the bottom!				
Angle 2: SC-6 Remainder View (**[g] [÷R]**)	Key **23 [ENTER] 4 [g] [÷R]**.	Floor 1 (**X**) shows **5** (Full Rows / Quotient) and Floor 2 (**Y**) shows **3** (Leftover Tiles / Remainder) at the exact same time!				
Angle 3: 3-Click Self-Proof (**[f] [LASTX]**)	Right after **[g] [÷R]**, press **[f] [LASTX] [×] [+]**.	**[f] [LASTX]** brings back the divisor **4**, **[×]** multiplies 5 × 4 = 20 while Remainder **3** waits on Floor 2, and **[+]** adds 20 + 3 = 23!				
Step	Keystroke	Floor 4 (**T**)	Floor 3 (**Z**)	Floor 2 (**Y**)	Floor 1 (**X**)	What Just Happened?
:---:	:---	:---:	:---:	:---:	:---:	:---
1	**135 [ENTER]**	**0**	**0**	**135**	**135**	Push 135 bearings onto Floor 2 (**Y**).
2	**8**	**0**	**0**	**135**	**8**	Type 8 bearings per skateboard on Floor 1 (**X**).
3	**[g] [÷R]**	**0**	**0**	**7**	**16**	BOOM! **X = 16** full boards, **Y = 7** leftover bearings!
4	**[f] [LASTX]**	**0**	**7**	**16**	**8**	Pulls divisor **8** back from memory! Remainder **7** rides up to **Z**.
5	**[×]**	**0**	**0**	**7**	**128**	Multiplies 16 × 8 = 128 bearings used. Leftover **7** drops to **Y**.
6	**[+]**	**0**	**0**	**0**	**135**	Adds 128 + 7 = 135. Matches starting number—100% Proven!

Problem 1 (Warm-Up): Arrange 31 Counting Tiles into groups of 6. How many full groups of 6 can you make, and how many tiles are left over?

1. [BLOCKS] TOUCH IT: `[ ]` Built 5 rows of 6 with 1 leftover tile.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (`31 [ENTER] 6 [g] [÷R]`): Snapshot `Y (Remainder): 1` | `X (Quotient): 5`	3. [PROVE] VERIFIED SOLUTION: Screen returns to `31`! Final Answer: `5 R 1` (or 51/6).

Problem 2 (Multiple-Ways Target): A 3D printer spool has 250 grams of filament. Each Prime Studio box lid takes 36 grams. Find (a) how many full lids you can print, (b) how many grams are left over, and (c) prove it with `[f] [LASTX]`.

1. [BLOCKS] TOUCH IT: `[ ]` Modeled 36g chunks out of 250g.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (`250 [ENTER] 36 [g] [÷R]`): Snapshot `Y (Leftover g): 34` | `X (Full Lids): 6`	3. [PROVE] VERIFIED SOLUTION: 6 × 36 = 216 -> 216 + 34 = 250. Final Answer: `6 lids, 34g left over`.

Problem 3 (Boss Level — "Do We Need an Extra Van?"): 115 middle schoolers are going to a robotics tournament. Each van holds 14 students. How many vans must the school book so nobody gets left behind?

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (`115 [ENTER] 14 [g] [÷R]`): Snapshot `Y (Students Left): 3` | `X (Full Vans): 8` • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (`[STACK-HACK] [g] [÷R]` Key): `115 [ENTER] 14 [g] [÷R]` -> `8 R 3` (so 9 vans). Way B (`[SPARK]` Multiply-Up Table): `14 [ENTER] 8 [×]` = `112` (3 kids left!) vs `14 [ENTER] 9 [×]` = `126` (all 115 fit -> 9 vans)!	3. [PROVE] VERIFIED SOLUTION: 8 vans hold 8 × 14 = 112 kids, leaving `3` kids standing in the parking lot! So you need 9 vans total.

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (`[STACK-HACK] [g] [÷R]` Key): `115 [ENTER] 14 [g] [÷R]` -> `8 R 3` (so 9 vans). Way B (`[SPARK]` Multiply-Up Table): `14 [ENTER] 8 [×]` = `112` (3 kids left!) vs `14 [ENTER] 9 [×]` = `126` (all 115 fit -> 9 vans)!

Day 2 (Lesson 1.2): Powers & Exponents — Squares, Cubes, and Tower Growth

`SPIRAL PATH: Thread C (Scale & Growth) • [[BUILDER] VISIT 1: BUILDER WAY — 2D/3D Tile Arrays -> REP Ladder] // VARIETY MODE: [DETECTIVE] Mode A: Spot the Glitch ('2^3 = 6 or 8?')`

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Leo says 2^3 = 6. Snap a 2 × 2 × 2 cube of blocks (8 blocks!) and press `2 [x³]` (`[g] [x²]`). Screen says `8`! You just busted the #1 6th-grade math myth in 15 seconds! (`[ ] +5 Spark XP!`)

The Real Question: Why does 3^4 mean 3 × 3 × 3 × 3 = 81 instead of 3 × 4 = 12—and how can you see the difference with physical tiles in 5 seconds?

Why Should I Care? (2-Min Hook): The #1 trap in 6th-grade math is seeing 2^3 and writing 6. If you fold a piece of paper in half 3 times (2^3), you don't get 6 layers—you get 2 × 2 × 2 = 8 layers! Exponents are how viral videos, bacteria, and Minecraft slime splits explode in size.
Why Try a New Way Today? (When the Old Way Gets Annoying): Building a 2D square (4^2=16) or 3D cube (4^3=64) with physical tiles (`[BUILDER]`) cures the '4^3=12' trap forever, while the `[g] [REP]` ladder (`[SPARK]`) and `[g] [y⊠]` (`[STACK-HACK]`) take over when powers grow too huge for your desk!

Camera Angle	What You Do	What You See
Angle 1: Physical Blocks (`COUNT-01`)	Build a 4 by 4 flat square of Counting Tiles (4^2 = 16). Now stack 4 of those square layers high (4^3 = 64).	A 2D square is `[x²]` (2 dimensions: length × width). A 3D cube is `[x³]` (3 dimensions: length × width × height)!
Angle 2: SC-6 Dedicated Power Keys	For squares: type `4 [x²]` -> `16`. For cubes: type `4 [x³]` -> `64`. For any power b^n: type `4 [ENTER] 3 [g] [y⊠]` -> `64`.	Floor 2 (`Y`) is the Base (what tile size you're using). Floor 1 (`X`) is the Exponent (how many dimensions/times you multiply it).
Angle 3: SC-6 Constant `[g] [REP]` Ladder	Type `4 [g] [REP]` (fills all 4 floors with `4`). Press `[×]` once -> `16` (4^2). Press `[×]` again -> `64` (4^3). Press `[×]` again -> `256` (4^4)!	You literally feel each `[×]` click multiply by another `4`—proving 4^3 is three 4s multiplied, never 4 × 3!

Problem 1 (Warm-Up Square vs. Double): Build a 5 × 5 square array of tiles. Find 5^2 and compare it to 5 × 2.

1. [BLOCKS] TOUCH IT: `[ ]` Built 5 × 5 square (25 tiles, NOT 10 tiles!).	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (`5 [x²]`): Snapshot `X: 25` (whereas `5 [ENTER] 2 [×]` is `10`).	3. [PROVE] VERIFIED SOLUTION: Press `[f] [√x]` (`[f] [x²]`) and watch `25` shrink right back to side length `5`!

Problem 2 (Multiple-Ways Target): Write 2^5 as repeated multiplication and find its value using both the `[g] [REP]` ladder and `[g] [y⊠]`.

1. [BLOCKS] TOUCH IT: 2 × 2 × 2 × 2 × 2 (5 factors of 2).	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (`2 [ENTER] 5 [g] [y⊠]`): Before `[g] [y⊠]`, `Y (Base): 2` | `X (Exponent): 5`. Result: `32`.	3. [PROVE] VERIFIED SOLUTION: Steps through `4 -> 8 -> 16 -> 32`!

Problem 3 (Boss Level — Perfect Cube Mystery): A giant Rubik's cube is made of 216 small cubes. How many small cubes wide is one edge? (Test with [x³] or [g] [√y]: 216 [ENTER] 3 [g] [√y]).		
1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**6 [x³]**): **X: 216**. Final Answer: **6 cubes wide** (6^3 = 216). • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[SPARK] [g] [REP]** Cube Ladder): **6 [g] [REP] [×] [×]** = **216** (edge = 6). Way B (**[STACK-HACK]** Cube Root **[g] [√y]**): **216 [ENTER] 3 [g] [√y]** = 6!	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.
[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[SPARK] [g] [REP]** Cube Ladder): **6 [g] [REP] [×] [×]** = **216** (edge = 6). Way B (**[STACK-HACK]** Cube Root **[g] [√y]**): **216 [ENTER] 3 [g] [√y]** = 6!		

Day 3 (Lesson 1.3): Order of Operations Without Parentheses Torture

SPIRAL PATH: Thread D (Inside-Out Chunking) • [[BUILDER] VISIT 1: BUILDER WAY — Launch of Thread D] // VARIETY MODE: [TARGET] Mode B: Mystery Target (2-Room Elevator)

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Put 5 tiles on the left of your desk and 2 rows of 3 tiles (6) on the right. Total is 11! Now type **5 [ENTER] 2 [ENTER] 3 [×]**—see **Y: 5** waiting while **X: 6** finishes! Press **[+]** -> **11**! (**[ ] +5 Spark XP!**)

The Real Question: How can you solve a huge multi-step math problem without copying 6 lines of scratch work or losing points to parentheses errors?

Why Should I Care? (2-Min Hook): Why do kids hate Order of Operations? Because your teacher makes you copy the whole line over and over, and if you type it into a regular calculator, forgetting one parenthesis gives the wrong answer. On the SC-6, there are no parentheses keys at all—because the 4-story glass elevator (X, Y, Z, T) automatically holds your pieces while you work from the inside out!
Why Try a New Way Today? (When the Old Way Gets Annoying): Rewriting 5 lines of PEMDAS by hand causes copying mistakes. Grouping tiles into separate 'rooms' (**[BUILDER]**) and solving inside-out on the 4-line elevator (**[STACK-HACK]**) lets Floor 2 (**Y**) hold Room 1 while you solve Room 2 on Floor 1 (**X**)!

Textbook Multi-Line Rewrite (Old Way)	SC-6 Elevator Snapshot (Our Way — Write 2 Numbers!)				
Line 1: (8 + 4) × (9 - 3) Line 2: = 12 × (9 - 3) Line 3: = 12 × 6 Line 4: = 72	1. Solve Room 1: **8 [ENTER] 4 [+]** -> **12** waits on the elevator. 2. Solve Room 2: **9 [ENTER] 3 [-]** -> **6** appears on Floor 1 (**X**), while **12** sits right above it on Floor 2 (**Y**)! 3. Write Snapshot: **Y: 12 | X: 6** 4. Press **[×]** -> **72**. Done!				
Step	Keystroke	Floor 3 (**Z**)	Floor 2 (**Y**)	Floor 1 (**X**)	Inside-Out Game Plan
:---:	:---	:---:	:---:	:---:	:---
1	**40 [ENTER]**	**0**	**40**	**40**	Park **40** upstairs on the balcony to wait.
2	**3 [ENTER]**	**40**	**3**	**3**	Park **3** on Floor 2 (**40** rides up to Floor 3 **Z**!).
3	**2 [ENTER] 3 [+]**	**40**	**3**	**5**	Solve inside parentheses (2+3) = 5.
4	**[x²]**	**40**	**3**	**25**	Square it: 5^2 = 25. Notice **3** is still waiting on **Y**!
5	**[×]**	**0**	**40**	**75**	Multiply 3 × 25 = 75. Notice **40** drops down to **Y**!
6	**5 [÷]**	**0**	**40**	**15**	Divide 75 ÷ 5 = 15. Snapshot before final key: **Y: 40 | X: 15**.
7	**[-]**	**0**	**0**	**25**	Subtract 40 - 15 = 25! Zero scratch-paper rewrites!

Problem 1 (Warm-Up Two-Room Combo): Evaluate (14 + 6) ÷ (7 - 2).

1. [BLOCKS] TOUCH IT: **[ ]** Built Group 1 (14+6=20 tiles) and Group 2 (7-2=5 tiles).	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**14 [ENTER] 6 [+] 7 [ENTER] 2 [-]**): Snapshot before **[÷]**: **Y: 20 | X: 5**	3. [PROVE] VERIFIED SOLUTION: Result **4** (**4 × 5 = 20**). Final Answer: **4**.

Problem 2 (Multiple-Ways Target): Evaluate 60 - 4 × (3 + 2)^2.		
1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**60 [ENTER] 4 [ENTER] 3 [ENTER] 2 [+] [x²] [×]**): Snapshot before **[-]**: **Y: 60 | X: 100**... wait: if the problem is 120 - 4 × (3 + 2)^2, let's do 120 - 4 × (3+2)^2: **120 [ENTER] 4 [ENTER] 3 [ENTER] 2 [+] [x²] [×]** gives **Y: 120 | X: 100**.	3. [PROVE] VERIFIED SOLUTION: **120 - 100 = 20**. Final Answer: **20**.

Problem 3 (Boss Level — Fraction Bar Grouping): Evaluate 3^3 + 9/15 - 3 × 2. (Hint: A fraction bar is just Top Room divided by Bottom Room!)		
1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: Top Room: **3 [x³] 9 [+]** (-> 36). Bottom Room: **15 [ENTER] 3 [ENTER] 2 [×] [-]** (-> 9). Snapshot before **[÷]**: **Y: 36 | X: 9**. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[STACK-HACK]** Inside-Out 4-Stack): **10 [ENTER] 15 [ENTER] 3 [-] [x²] 9 [÷] [+]** = 26. Way B (**[BUILDER]** 2-Room Chunking): Room 1 = **10**, Room 2 = **144 ÷ 9 = 16** -> **10 + 16** = 26!	3. [PROVE] VERIFIED SOLUTION: Final Answer: **4**.
[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[STACK-HACK]** Inside-Out 4-Stack): **10 [ENTER] 15 [ENTER] 3 [-] [x²] 9 [÷] [+]** = 26. Way B (**[BUILDER]** 2-Room Chunking): Room 1 = **10**, Room 2 = **144 ÷ 9 = 16** -> **10 + 16** = 26!		

Day 4 (Lesson 1.4): Prime Factorization — Why Factor Trees Are Just Stacked Blocks

SPIRAL PATH: Thread A (Factor & Split) • [[SPARK] VISIT 2: PATTERN WAY — Revisits Day 1 Tile Splitting] // VARIETY MODE: [BOOMERANG] Mode D: 2-Second Shortcut Race ({{key:FACT}})

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Stack a red **2** block (**12.6mm**), another **2** block, and a green **3** block (**20.0mm**). Their height matches the **12** tower (2 × 2 × 3 = 12)! Now type **12 [ENTER] [g] [FACT]** -> screen says **2^2 × 3**! (**[ ] +5 Spark XP!**)

The Real Question: What makes a number "Prime," and why do you always get the exact same prime blocks at the bottom of a factor tree no matter how you split the number first?

Why Should I Care? (2-Min Hook): Kids hate drawing factor trees because if your friend splits 60 into 6 × 10 and you split 60 into 4 × 15, your trees look completely different in the middle—and you worry yours is wrong! With the Prime Factor Studio (**PRIME-01**), you can physically see and hold the "atomic LEGO bricks" of any number.
Why Try a New Way Today? (When the Old Way Gets Annoying): On Day 1 we split numbers into equal rows one divisor at a time. When you want the deepest 'DNA building blocks' of a number without drawing lopsided paper factor trees, snap Prime Towers (**[BUILDER]**) or press **[g] [FACT]** (**[STACK-HACK]**)!

Camera Angle	What You Do	What You See
Angle 1: Rectangular Array Test (**COUNT-01**)	Try to arrange 13 Counting Tiles into a rectangle vs. 12 Counting Tiles.	**13** can ONLY make a single skinny 1 × 13 line (Prime!). **12** makes 2 × 6 and 3 × 4 rectangles (Composite!).
Angle 2: Prime Factor Tower (**PRIME-01**)	For **60**, grab two Blue **[2]** blocks, one Green **[3]** block, and one Yellow **[5]** block. Stack them in a socket on the lid!	Because the block heights are logarithmic (h = 42\log_{10}(p) mm), stacking **[2] + [2] + [3] + [5]** reaches exactly the 60 height mark! Group **[2][3]=6** and **[2][5]=10** (that's the 6 × 10 tree!), or group **[2][2]=4** and **[3][5]=15** (that's the 4 × 15 tree!). Same blocks!
Angle 3: SC-6 **[PRIME]** & **[g] [FACT]**	Key **60 [PRIME]** -> **NOT PRIME**. Now key **60 [g] [FACT]** (**[g] [PRIME]**).	The SC-6 screen immediately displays **2² × 3 × 5**!

Problem 1 (Prime Detective): Test 29 and 51—which one is secretly composite?		
1. [BLOCKS] TOUCH IT: Can 51 tiles split into 3 equal rows?	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: **29 [PRIME]** -> **PRIME**. **51 [PRIME]** -> **NOT PRIME**! Now press **51 [g] [FACT]** -> **3 × 17**.	3. [PROVE] VERIFIED SOLUTION: Screen shows **51**!

Problem 2 (Two Trees, One Tower): Build the Prime Factor Tower for 72. Show how both 8 × 9 and 6 × 12 use the exact same prime blocks.		
1. [BLOCKS] TOUCH IT: **[ ]** Stacked three **[2]** blocks (=8) and two **[3]** blocks (=9). Rearranged into ([2][3]) × ([2][2][3]) = 6 × 12.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**72 [g] [FACT]**): Screen displays **2³ × 3²**.	3. [PROVE] VERIFIED SOLUTION: Snapshot **Y: 8 | X: 9** -> **72**!

Problem 3 (Boss Level — Mystery Tower): What number has the prime factorization 2^2 × 3 × 7? Build it on the stack first, then verify with `[g] [FACT]`.

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (`2 [x²] 3 [×] 7 [×]`): Snapshot `Y: 12 | X: 7` -> **84**. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (`[SPARK]` Split-in-Chunks Tree): 180 = 18 × 10 = (2 · 3^2) × (2 · 5) = 2^2 · 3^2 · 5. Way B (`[STACK-HACK]` SC-6 `[g] [FACT]` Key): `180 [ENTER] [g] [FACT]` -> `2^2 × 3^2 × 5`!	3. [PROVE] VERIFIED SOLUTION: **84** decomposes right back into `2² × 3 × 7`!

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (`[SPARK]` Split-in-Chunks Tree): 180 = 18 × 10 = (2 · 3^2) × (2 · 5) = 2^2 · 3^2 · 5. Way B (`[STACK-HACK]` SC-6 `[g] [FACT]` Key): `180 [ENTER] [g] [FACT]` -> `2^2 × 3^2 × 5`!

Day 5 (Lesson 1.5): GCF & LCM in One Look — Lists, Trees, and Venn Diagrams

SPIRAL PATH: Thread A (Factor & Split) • [[STACK-HACK] VISIT 3: STACK HACKER WAY — Boss Revisit of Days 1 & 4] // VARIETY MODE: DICE Mode E: Speed Pack Challenge ({{key:GCF}} & {{key:LCM}})

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Stand the Prime Towers for **12** (2,2,3) and **18** (2,3,3) next to each other. Both share one **2** and one **3** (2 × 3 = 6)! Now press `12 [ENTER] 18 [GCF]` -> **6** in 1 second! (`[ ] +5 Spark XP!`)

The Real Question: Why does the textbook make you find GCF and LCM using lists, trees, AND Venn diagrams—and how do two side-by-side block towers show both GCF and LCM at the same time?

Why Should I Care? (2-Min Hook): Usually, GCF and LCM take half a page of tedious factor lists and multiple lists, and kids constantly mix up which one is which. When you stand two Prime Factor Studio towers next to each other on the 12-socket lid, GCF is the blocks they share in common (the overlap), and LCM is the tallest tower needed to cover both!
Why Try a New Way Today? (When the Old Way Gets Annoying): Listing every factor by hand works for **6** and **8**, but for **36** and **48** you'll miss factors and waste 10 minutes! Laying Day 4's Prime Towers side-by-side (`[BUILDER]`) shows why GCF is the shared overlap, and `[GCF]`/`[f] [LCM]` (`[STACK-HACK]`) solves it in 2 clicks!

Textbook Method	How Your 1 Block Build (PRIME-04) Shows It	SC-6 Instant Keystroke		
1. Prime Factor Towers	Socket A (**20**): `[2] + [2] + [5]` Socket B (**50**): `[2] + [5] + [5]`	`20 [g] [FACT]` -> `2² × 5` `50 [g] [FACT]` -> `2 × 5²`		
2. Venn Overlap (GCF)	Look across the two towers: both share one `[2]` and one `[5]`. Stack `[2] + [5]` in the middle socket = 10!	`20 [ENTER] 50 [GCF]` -> **10** (Or `[f] [C.FAC] [f] [PRIME]` to list all common factors!)		
3. Venn Union (LCM)	Take the max count of each block color: two `[2]`s and two `[5]`s. Stack `[2]+[2]+[5]+[5]` = 100!	`20 [ENTER] 50 [f] [LCM]` (`[f] [GCF]`) -> **100**		
Step	Keystroke	Floor 2 (**Y**)	Floor 1 (**X**)	What Just Happened?
:---:	:---	:---:	:---:	:---
1	`12 [ENTER] 18 [GCF]`	**0**	**6**	Greatest Common Factor of 12 and 18 is **6**.
2	`12 [ENTER] 18 [f] [LCM]`	**6**	**36**	Least Common Multiple is **36** (while GCF **6** waits on **Y**!).
3	`[×]`	**0**	**216**	GCF × LCM = 6 × 36 = 216 (Matches 12 × 18 = 216!).

Problem 1 (Warm-Up Twin Towers): Build the Prime Factor Towers for 18 (`[2] [3] [3]`) and 24 (`[2] [2] [2] [3]`). Find both the GCF and LCM.

1. [BLOCKS] TOUCH IT: `[ ]` Shared overlap: one `[2]` and one `[3]` (= 6). Tallest union: three `[2]`s and two `[3]`s (= 72).	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: `18 [ENTER] 24 [GCF]` -> **6**. `18 [ENTER] 24 [f] [LCM]` -> **72**. Snapshot `Y: 6 | X: 72`.	3. [PROVE] VERIFIED SOLUTION: 6 × 72 = 432 (and 18 × 24 = 432!). Final Answer: `GCF = 6, LCM = 72`.

Problem 2 (Hot Dog & Bun Synchronizer): Hot dogs come in packs of 8 (`[2] [2] [2]`) and buns come in packs of 12 (`[2] [2] [3]`). What is the smallest number of hot dogs and buns you can buy to have zero leftovers?		
1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**`8 [ENTER] 12 [f] [LCM]`**): Snapshot **`Y: 8 | X: 12`** -> **`24`**.	3. [PROVE] VERIFIED SOLUTION: 24 ÷ 8 = 3 packs of hot dogs; 24 ÷ 12 = 2 packs of buns!

Problem 3 (Boss Level — Kit Packaging): You have 36 red counters and 48 yellow counters to pack into identical station bags with none left over. What is the greatest number of bags you can make, and how many of each color go in a bag?		
1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**`36 [ENTER] 48 [GCF]`**): Snapshot **`Y: 36 | X: 48`** -> **`12`** bags. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**`[SPARK] [g] [REP]`** Skip-Count Race): Step **`12 [g] [REP] [+]...`** (**`12, 24, 36, 48, 60`**) vs **`15 [g] [REP] [+]...`** (**`15, 30, 45, 60`**). Way B (**`[STACK-HACK] [f] [LCM]`** Key): **`12 [ENTER] 15 [f] [LCM]`** = 60!	3. [PROVE] VERIFIED SOLUTION: **`36 [ENTER] 12 [÷]`** -> **`3 red`**; **`48 [ENTER] 12 [÷]`** -> **`4 yellow`** per bag!
[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**`[SPARK] [g] [REP]`** Skip-Count Race): Step **`12 [g] [REP] [+]...`** (**`12, 24, 36, 48, 60`**) vs **`15 [g] [REP] [+]...`** (**`15, 30, 45, 60`**). Way B (**`[STACK-HACK] [f] [LCM]`** Key): **`12 [ENTER] 15 [f] [LCM]`** = 60!		

CHAPTER 2: Fractions & Decimals (Days 6–10)

Aligned to Big Ideas Math Grade 6, Chapter 2: Equivalence, Multiplying Fractions, Dividing Fractions, and All Four Decimal Operations.

Day 6 (Lesson 2.1): Equivalent Fractions, Mixed Numbers & Improper Fractions — Same Height, Different Outfits

SPIRAL PATH: Thread B (Flip & Inverse) + Thread A Boomerang • [[BUILDER] VISIT 1: BUILDER WAY — Revisits Day 5 GCF] // VARIETY MODE: [SPARK] Mode C: Same-Height Tower Match ({{key:SIMP}})

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Hold one **1/2** tower (**42mm**) next to three **1/6** blocks (**3 × 14mm = 42mm**). Exact same height! Now type **[.] 3 [.] 6 [ENTER] [g] [SIMP]**—watch **3/6** snap into **1/2**! (**[ ] +5 Spark XP!**)

The Real Question: Why does 12/18 equal 2/3, and why are 13/4, 7/4, and 1.75 literally the exact same height?

Why Should I Care? (2-Min Hook): Fractions are where most middle schoolers decide they hate math—because teachers ask you to reduce fractions, convert mixed numbers to improper fractions, AND convert them to decimals, making it feel like three unrelated rules. In the Fraction Studio (**FRAC-01**), every single block is engineered to an 84.0 mm Whole. When two towers stand at the exact same millimeter height, they are equal—period!
Why Try a New Way Today? (When the Old Way Gets Annoying): In Chapter 1 (**Day 5**), you learned **[GCF]**. Today that exact skill boomerangs back! Simplifying a fraction is just dividing top and bottom by their GCF—or matching 84mm Fraction Towers (**[BUILDER]**) side-by-side and pressing **[g] [SIMP]** (**[STACK-HACK]**)!

Textbook Requirement	Physical Block View (FRAC-01)	SC-6 1-Click Costume Change
1. Step-by-Step Simplification (12/18 -> 2/3)	Stand four 1/6 blocks (4/6 = 56 mm) next to two 1/3 blocks (2/3 = 56 mm). Same height!	Enter **[.] 12 [.] 18 [ENTER]**. Press **[g] [SIMP]** (**[g][a b/c]**) once -> **6/9** (divides by 2!). Press **[g] [SIMP]** again -> **2/3** (divides by 3!).
2. Mixed Number <-> Improper (13/4 <-> 7/4)	Stand one **1 Whole** (84 mm) + three **1/4** blocks (63 mm) next to seven **1/4** blocks (147 mm).	Enter **1 [.] 3 [.] 4 [ENTER]**. Press **[f] [d/c]** (**[f][a b/c]**) -> **7/4**. Press **[a b/c]** -> **1 3/4**!
3. Fraction <-> Decimal (13/4 <-> 1.75)	Compare the tower against the 100-grid scale.	Press **[FDISP]** to toggle the display between **1 3/4** and **1.75** instantly!

Problem 1 (Step-by-Step Simplifier): Simplify 24/36 step-by-step and write the intermediate fraction you pass along the way.

1. [BLOCKS] TOUCH IT: **[ ]** Matched height to 2/3 tower (56.0 mm).	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**[.] 24 [.] 36 [ENTER] [g] [SIMP]**): First **[g] [SIMP]** press shows **12/18**, next shows **6/9**, final shows **2/3**.	3. [PROVE] VERIFIED SOLUTION: Toggles cleanly to **0.6667** (2/3).

Problem 2 (The 3-Outfit Switcher): Convert 19/8 into (a) a Mixed Number and (b) a Decimal.

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**[.] 19 [.] 8 [ENTER]**): Press **[a b/c]** -> **2 3/8**. Press **[FDISP]** -> **2.375**.	3. [PROVE] VERIFIED SOLUTION: Shows Quotient **X: 2** and Remainder **Y: 3** (2 R 3 = 23/8)!

Problem 3 (Boss Level — Adding Unlike Towers): Build 1/3 + 1/4 on the 12-socket Fraction Lid Tray and verify on SC-6.

1. [BLOCKS] TOUCH IT: **[ ]** 28 mm (1/3) + 21 mm (1/4) = 49 mm (matches seven 1/12 blocks!).	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**[.] 1 [.] 3 [ENTER] [.] 1 [.] 4 [+]**): Snapshot **Y: 1/3 | X: 1/4** -> **7/12**. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[BUILDER]** 84mm Tower Swap): Swap 1/3 (**28mm**) for four 1/12 blocks and 1/4 (**21mm**) for three 1/12 blocks -> **7/12**. Way B (**[STACK-HACK]** SC-6 Fraction Stack): **[.] 1 [.] 3 [ENTER] [.] 1 [.] 4 [+]** = **7/12**!	3. [PROVE] VERIFIED SOLUTION: Subtracts **1/4** and returns to **1/3**!

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[BUILDER]** 84mm Tower Swap): Swap 1/3 (**28mm**) for four 1/12 blocks and 1/4 (**21mm**) for three 1/12 blocks -> **7/12**. Way B (**[STACK-HACK]** SC-6 Fraction Stack): **[.] 1 [.] 3 [ENTER] [.] 1 [.] 4 [+]** = **7/12**!

Day 7 (Lesson 2.2): Multiplying Fractions & Mixed Numbers — Taking a Fraction OF a Fraction

SPIRAL PATH: Thread B (Flip & Inverse) • [[SPARK] VISIT 2: PATTERN WAY — Revisits Day 6 Fraction Towers] // VARIETY MODE: [DETECTIVE] Mode A: Prediction Bet ('Does × Make It Bigger or Smaller?')

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Take two **1/3** blocks (2/3). Hand half of them (1/2 of 2/3) to a friend—you're holding just one **1/3** block! Key **[.] 1 [.] 2 [ENTER] [.] 2 [.] 3 [×]** -> **1/3**! (**[ ] +5 Spark XP!**)

The Real Question: When you multiply two proper fractions, why does the answer get SMALLER instead of bigger?

Why Should I Care? (2-Min Hook): In elementary school, multiplying always made numbers bigger. So when a 6th grader sees 1/2 × 3/4 = 3/8, their brain says "Wait, 3/8 is smaller than 3/4!" Replace the **×** symbol with the word "OF": if there is 3/4 of a pizza left in the fridge and you eat 1/2 OF it, of course your slice is smaller than what was in the box!
Why Try a New Way Today? (When the Old Way Gets Annoying): Adding fractions (**Day 6**) required common denominators. Multiplying fractions is way faster once you see with 84mm blocks (**[BUILDER]**) that '1/2 of 2/3' just means taking 1 of the 2 thirds—and cross-canceling (**[SPARK]**) shrinks big numbers before you even multiply!

Camera Angle	What You Do	What You See
Angle 1: Physical Fraction Towers (**FRAC-01**)	Place three **1/4** blocks on the desk (3/4 = 63 mm). Now find the tower that is half as tall (31.5 mm).	Three **1/8** blocks (3/8) are exactly half the height of 3/4!
Angle 2: Mixed Numbers on Stack	For 11/2 × 22/3, enter **1 [.] 1 [.] 2 [ENTER] 2 [.] 2 [.] 3**.	Before pressing **[×]**, tap **[f] [d/c]** if your teacher wants to see the improper fractions (**Y: 3/2 | X: 8/3**), then press **[×]** -> **4**!

Problem 1 (Half of a Fraction): Find 2/3 × 3/4.

1. [BLOCKS] TOUCH IT: **[ ]** Took 2/3 of a 3/4 tower (-> 1/2 of a whole, 42 mm).	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**[.] 2 [.] 3 [ENTER] [.] 3 [.] 4 [×]**): Snapshot **Y: 2/3 | X: 3/4** -> **1/2**.	3. [PROVE] VERIFIED SOLUTION: Divides by **3/4** and restores **2/3**!

Problem 2 (Mixed Number Recipe Scaling): A batch of slime uses 13/4 cups of glue. You are making 21/2 batches. How much glue do you need?

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**1 [.] 3 [.] 4 [ENTER] 2 [.] 1 [.] 2**): Snapshot **Y: 1 3/4 (7/4) | X: 2 1/2 (5/2)**. Press **[×]** -> **4 3/8** (or **35/8** via **[f] [d/c]**).	3. [PROVE] VERIFIED SOLUTION: Restores **1 3/4**!

Problem 3 (Boss Level — Triple Fraction Chain): Evaluate 4/5 × 5/6 × 3/4.

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**[.] 4 [.] 5 [ENTER] [.] 5 [.] 6 [×] [.] 3 [.] 4 [×]**): Snapshot before second **[×]**: **Y: 2/3 | X: 3/4** -> **1/2**. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[SPARK]** Cross-Cancel Pattern): In 4/5 × 5/6 × 3/4, cancel the **5**s and **4**s first -> 3/6 = **1/2**. Way B (**[STACK-HACK]** Direct Stack Multiply): Multiply all three on the SC-6 -> **1/2**!	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[SPARK]** Cross-Cancel Pattern): In 4/5 × 5/6 × 3/4, cancel the **5**s and **4**s first -> 3/6 = **1/2**. Way B (**[STACK-HACK]** Direct Stack Multiply): Multiply all three on the SC-6 -> **1/2**!

Day 8 (Lesson 2.3): Dividing Fractions — Why "Keep-Change-Flip" Works Physically

SPIRAL PATH: Thread B (Flip & Inverse) • [[STACK-HACK] VISIT 3: STACK HACKER WAY — Boss Revisit of Days 1, 6 & 7] // VARIETY MODE: [BOOMERANG] Mode D: 2-Second Shortcut Race ({{key:1/x}} Flip)

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): How many **1/4** blocks fit next to one **1/2** block? Lay them down: 2 fit! Now type **[.] 1 [.] 2 [ENTER] [.] 1 [.] 4 [÷]** -> **2**! Dividing by 1/4 is the same as multiplying by 4! (**[ ] +5 Spark XP!**)

The Real Question: Why does dividing by a fraction mean multiplying by its upside-down reciprocal (**[f] [1/x]**)—and why do Common Denominators and Reciprocals always give the exact same answer?

Why Should I Care? (2-Min Hook): Every middle schooler has heard "Keep, Change, Flip!"—and then immediately forgets which fraction to flip, or why on earth you're flipping anything! With the 84 mm Fraction Towers, 3/4 ÷ 1/8 simply asks: "How many 1/8 blocks fit alongside a 3/4 tower?"
Why Try a New Way Today? (When the Old Way Gets Annoying): On small problems like 3/4 ÷ 1/8, laying 1/8 blocks (**[BUILDER]**) along a 3/4 bar is super clear (6 fit!). But for 63/4 ÷ 3/4, you don't want to count 27 plastic fourths—flipping the divisor with **[f] [1/x]** (**[STACK-HACK]**) turns division into Day 7's fast multiplication!

Method	How It Works on Your Desk & SC-6			
1. Physical Tower Fit (**FRAC-01**)	Stand a 3/4 tower (63 mm) in Socket 1. Stack 1/8 blocks (10.5 mm each) in Socket 2 until the heights match. It takes 6 blocks!			
2. Common Denominator View	Since 3/4 = 6/8, the problem is \frac{6 eighths}{1 eighth} = 6 ÷ 1 = 6.			
3. SC-6 Reciprocal Key (**[f] [1/x]**)	Key **[.] 3 [.] 4 [ENTER] [.] 1 [.] 8**. Now press **[f] [1/x]** (**[f][FDISP]**)—watch **1/8** flip upside-down into **8** right on Floor 1 (**X**)! Press **[×]** -> **6**! (Or press **[÷]** directly to get **6**!)			
Step	Keystroke	Floor 2 (**Y**)	Floor 1 (**X**)	What Just Happened?
:---:	:---	:---:	:---:	:---
1	**[.] 3 [.] 4 [ENTER]**	**3/4**	**3/4**	Dividend 3/4 rides up to Floor 2 (**Y**).
2	**[.] 2 [.] 5**	**3/4**	**2/5**	Divisor 2/5 enters Floor 1 (**X**).
3	**[f] [1/x]** (**[f][FDISP]**)	**3/4**	**5/2**	Flip! The divisor on Floor 1 flips into its reciprocal **5/2**!
4	**[×]**	**0**	**1 7/8**	Multiplies 3/4 × 5/2 = 15/8 = 17/8!

Problem 1 (Tower Fit Check): How many 1/6 blocks fit inside a 2/3 tower? Solve 2/3 ÷ 1/6.		
1. [BLOCKS] TOUCH IT: **[ ]** Four 1/6 blocks (56 mm) match two 1/3 blocks (56 mm).	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**[.] 2 [.] 3 [ENTER] [.] 1 [.] 6 [f] [1/x]**): Snapshot before **[×]**: **Y: 2/3 | X: 6** -> **4**.	3. [PROVE] VERIFIED SOLUTION: 4 × 1/6 = 4/6 = 2/3!

Problem 2 (Reciprocal Flip): Solve 5/6 ÷ 5/12 using [f] [1/x] and write the flipped reciprocal snapshot.		
1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**[.] 5 [.] 6 [ENTER] [.] 5 [.] 12 [f] [1/x]**): Snapshot before **[×]**: **Y: 5/6 | X: 12/5**. Press **[×]** -> **2**.	3. [PROVE] VERIFIED SOLUTION: Returns **5/6**!

Problem 3 (Boss Level — Mixed Number Board Cutting): You have a board that is 63/4 feet long. You are cutting it into shelves that are 11/8 feet long each. How many shelves can you cut?

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**`6 [.] 3 [.] 4 [ENTER] 1 [.] 1 [.] 8 [÷]`**): Snapshot **`Y: 6 3/4 (27/4) | X: 1 1/8 (9/8)`** -> **6** shelves. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**`[SPARK]`** Common-Denominator Fourths): 63/4 = 27/4, and 27 fourths ÷ 3 fourths = 9. Way B (**`[STACK-HACK]`** Reciprocal Flip **`[f] [1/x]`**): **`6 [.] 3 [.] 4 [ENTER] [.] 3 [.] 4 [f] [1/x] [×]`** = 9 shelves!	3. [PROVE] VERIFIED SOLUTION: 6 × 11/8 = 63/4!

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**`[SPARK]`** Common-Denominator Fourths): 63/4 = 27/4, and 27 fourths ÷ 3 fourths = 9. Way B (**`[STACK-HACK]`** Reciprocal Flip **`[f] [1/x]`**): **`6 [.] 3 [.] 4 [ENTER] [.] 3 [.] 4 [f] [1/x] [×]`** = 9 shelves!

Day 9 (Lesson 2.4): Adding, Subtracting & Multiplying Decimals Without Crooked Columns

`SPIRAL PATH: Thread B (Flip & Inverse) • [[SPARK] VISIT 2: PATTERN WAY — Revisits Day 6 Mixed Numbers] // VARIETY MODE: [TARGET] Mode B: Split the Whole & Decimal ({{key:IP}} & {{key:FP}})`

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Type **`14.75 [ENTER]`** and press **`[f] [IP]`** (**`[f][PV]`**)—it grabs the whole **14**! Press **`[f] [LASTX] [g] [FP]`** (**`[g][PV]`**)—it grabs the decimal **`.75`**! Zero crooked paper columns! (**`[ ] +5 Spark XP!`**)

The Real Question: How do you keep whole parts (**`[f] [IP]`**) and decimal parts (**`[g] [FP]`**) straight so you never put the decimal point in the wrong spot?

Why Should I Care? (2-Min Hook): On paper, if you compute 14.6 + 3.85 and line up the right edge instead of the decimal point, you get the whole problem wrong. And when you multiply 2.4 × 0.15, counting decimal jumps by hand is a magnet for silly mistakes. Let's use the Place Value Studio (**`PLACE-01`**) and the SC-6 **`[f] [IP]`** (Integer Part), **`[g] [FP]`** (Fractional Part), and **`[$]`** (Currency) keys to make decimal place value bulletproof!
Why Try a New Way Today? (When the Old Way Gets Annoying): Lining up decimal points by hand on lined paper is frustrating when columns drift. Splitting wholes and tenths with **`[f] [IP]`** and **`[g] [FP]`** (**`[BUILDER]`**) or entering decimals directly on the stack (**`[STACK-HACK]`**) keeps place value locked automatically!

Camera Angle	What You Do	What You See
Angle 1: **`[f] [IP]`** & **`[g] [FP]`** Split	Type **`14.85`**. Press **`[f] [IP]`** (**`[f][PV]`**) to see the whole part (**14**), or **`[g] [FP]`** (**`[g][PV]`**) to isolate the decimal part (**`0.85`**).	You never confuse dollars (whole units) with cents (hundredths)!
Angle 2: Place Value Regrouping (**`PLACE-01`**)	When tenths exceed 10 (e.g., 0.6 + 0.7 = 1.3), 10 tenths bundle into **1** whole unit on the carry tray.	Matches the stack sum **`0.6 [ENTER] 0.7 [+]`** -> **`1.30`**.
Angle 3: Instant **`[f] [LASTX]`** Check	After any decimal addition, subtraction, or multiplication, press **`[f] [LASTX]`** and the opposite operation!	If you added, **`[f] [LASTX] [-]`** restores the first number. If you multiplied, **`[f] [LASTX] [÷]`** restores it!

Problem 1 (Snack Bar Change Check): You have a $20.00 bill and buy a burger for $8.75 and a smoothie for $4.60. How much change do you get back?

1. [BLOCKS] TOUCH IT: **`[ ]`** Grouped dollars (8 + 4 = 12) and cents (0.75 + 0.60 = 1.35 -> 13.35$).	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**`20 [ENTER] 8.75 [ENTER] 4.60 [+]`**): Snapshot before **`[-]`**: **`Y: 20.00 | X: 13.35`**. Press **`[-]`** -> **`6.65`**.	3. [PROVE] VERIFIED SOLUTION: 6.65 + 13.35 = 20.00!

Problem 2 (Decimal Multiplication & Estimate): Find 3.6 × 2.25. First check the whole-number ballpark (4 × 2 = 8), so you know 810 or 0.81 would be a decimal-point mistake!

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**`3.6 [ENTER] 2.25 [×]`**): Snapshot **`Y: 3.60 | X: 2.25`** -> **`8.10`** (8.1).	3. [PROVE] VERIFIED SOLUTION: 8.1 ÷ 2.25 = 3.6!

Problem 3 (Boss Level — Fraction-Decimal Twin Check): Solve 4.75 × 1.2 as decimals, then press [FDISP] to see the exact fraction equivalent!

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**4.75 [ENTER] 1.2 [×]**): Result **5.70** (**5.7**). Press **[FDISP]** -> **5 7/10**! • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[STACK-HACK]** Decimal Stack): **4.75 [ENTER] 1.2 [×]** = **5.7**. Way B (**[SPARK]** Fraction Costume via **[FDISP]**): 43/4 × 11/5 = 19/4 × 6/5 = 57/10 = **5 7/10**!	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[STACK-HACK]** Decimal Stack): **4.75 [ENTER] 1.2 [×]** = **5.7**. Way B (**[SPARK]** Fraction Costume via **[FDISP]**): 43/4 × 11/5 = 19/4 × 6/5 = 57/10 = **5 7/10**!

Day 10 (Lesson 2.5): Dividing Decimals & The 3-Way Fraction-Decimal-Remainder Translator

SPIRAL PATH: Thread A & B Boomerang • [[BOOMERANG] VISIT 4: CROSS-CHAPTER BOOMERANG — Revisits Days 1, 6, 8 & 9] // VARIETY MODE: [BOOMERANG] Mode D: 1-Thumb Costume Change ({{key:FDISP}})

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Type **15 [ENTER] 4 [÷]** (**3.75**). Now tap **[FDISP]** three times: watch **3.75** morph into **3 3/4**, then **15/4**, then **3 R 3**! Four textbook answers with 1 thumb! (**[ ] +5 Spark XP!**)

The Real Question: Why are 15 ÷ 4 = 3 R 3, 33/4, and 3.75 all the exact same answer—and how do you divide by a decimal like 0.25?

Why Should I Care? (2-Min Hook): On paper, dividing 6.75 ÷ 0.25 requires sliding decimal points and long division houses. In real life, 0.25 is just a quarter (1/4)! Asking "What is 6.75 ÷ 0.25?" is literally asking "How many quarters are in $6.75?"
Why Try a New Way Today? (When the Old Way Gets Annoying): Moving the decimal scoop in long division is easy to mess up. Today connects 4 earlier days at once: decimal division (**Day 10**), fraction division (**Day 8**), mixed numbers (**Day 6**), and remainders (**Day 1**)—using **[FDISP]** to switch between them in 1 click!

Format	Keystrokes for 23 ÷ 4	What the Screen Shows
1. Quotient & Remainder	**23 [ENTER] 4 [g] [÷R]**	**Y: 3** (Remainder) and **X: 5** (Quotient) -> 5 R 3
2. Exact Decimal	**23 [ENTER] 4 [÷]**	**X: 5.75** -> 5.75
3. Mixed Fraction & Improper	Press **[FDISP]** (then **[f] [d/c]**)	**X: 5 3/4** (and **23/4**) -> 53/4

Problem 1 (The 3-Way Translator): Write 39 ÷ 8 in all three forms: (a) Quotient + Remainder, (b) Decimal, and (c) Mixed Number.

1. [BLOCKS] TOUCH IT: **[ ]** 4 full groups of 8 (=32) plus 7 eighths left over.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: **39 [ENTER] 8 [g] [÷R]** -> **4 R 7**. **39 [ENTER] 8 [÷]** -> **4.875** (press **[g] [SHOW] [g] [ENTER]** for full digits, or **[FDISP]** -> **4 7/8**).	3. [PROVE] VERIFIED SOLUTION: 4.875 × 8 = 39!

Problem 2 (How Many Quarters?): Solve 11.25 ÷ 0.25.

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**11.25 [ENTER] 0.25 [÷]**): Snapshot **Y: 11.25 | X: 0.25** -> **45**.	3. [PROVE] VERIFIED SOLUTION: 45 quarters × 0.25 = 11.25!

Problem 3 (Boss Level — Track Lap Split): A runner completes a 4.5-mile race in 0.75-mile laps, taking 6.3 minutes per lap. How many laps is the race, and what is the total time?

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**4.5 [ENTER] 0.75 [÷] 6.3 [×]**): Laps: **6**. Snapshot before **[×]**: **Y: 6 | X: 6.3** -> **37.8** minutes. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[STACK-HACK]** Decimal Division): **4.5 [ENTER] 0.75 [÷]** = 6 laps. Way B (**[BUILDER]** Fraction Tower Equivalent from Day 8): 41/2 ÷ 3/4 = 9/2 × 4/3 = 6 laps!	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[STACK-HACK]** Decimal Division): **4.5 [ENTER] 0.75 [÷]** = 6 laps. Way B (**[BUILDER]** Fraction Tower Equivalent from Day 8): 41/2 ÷ 3/4 = 9/2 × 4/3 = 6 laps!

CHAPTER 3: Ratios, Rates & Percents (Days 11–15)

Aligned to Big Ideas Math Grade 6, Chapters 3 & 4/5: Ratios, Tape Diagrams, Ratio Tables, Unit Rates, Percents, and Converting Measures.

Day 11 (Lesson 3.1): Ratios & Tape Diagrams — Part-to-Part vs. Part-to-Whole

SPIRAL PATH: Thread C (Scale & Ratio) • [[BUILDER] VISIT 1: BUILDER WAY — Revisits Day 6 Fraction Bars] // VARIETY MODE: [TARGET] Mode B: Mystery Box Weight (Tape Diagram)

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Put 2 Blue tiles and 3 Gold tiles on your desk (5 tiles total). If all 5 tiles weigh 20g, each tile is 4g! Now type **12 [ENTER] 18 [g] [a:b]** -> watch it simplify to **2:3**! (**[ ] +5 Spark XP!**)

The Real Question: When a problem gives you a ratio like 3:2, how do physical color tile bars keep you from mixing up "Part-to-Part" and "Part-to-Whole"?

Why Should I Care? (2-Min Hook): The Big Ideas textbook loves Tape Diagrams. Kids hate drawing them because their boxes come out uneven on paper. With the Counting & Array Tiles (**COUNT-01**), you don't draw boxes—you snap 3 Blue tiles and 2 Gold tiles onto your desk! Each tile is an equal "chunk" (box) of the tape diagram!
Why Try a New Way Today? (When the Old Way Gets Annoying): Students often confuse Part-to-Part (3:2) with Part-to-Whole (3/5). Building 2 rows of colored Counting Tiles (**[BUILDER]**) lets you physically see both the two parts AND the combined total bar before pressing **[g] [a:b]** (**[STACK-HACK]**)!

Textbook Representation	How Your 1 Tile Build (COUNT-01) Shows It	SC-6 Keystroke Flow
1. Tape Diagram	Lay a row of 3 Blue tiles above a row of 2 Gold tiles. Total tiles = 3 + 2 = 5 equal boxes!	**3 [ENTER] 2 [+]** -> **5** total boxes.
2. Value of 1 Box (Unit Scale)	Share **20** cups equally among the **5** tiles: each tile stands for 20 ÷ 5 = 4 cups!	**20 [ENTER] 5 [÷]** -> **4** cups per tile.
3. Equivalent Ratio (**[g] [a:b]**)	3 Blue tiles × 4 = 12 cups; 2 Gold tiles × 4 = 8 cups (12:8 = 3:2).	Key **12 [ENTER] 8 [g] [a:b]** (**[g][GCF]**) -> reduces ratio directly to **3:2**!

Problem 1 (Ratio Simplifier): A gaming clan has 18 builders and 12 speedrunners. Simplify the ratio of builders to speedrunners AND builders to total players.

1. [BLOCKS] TOUCH IT: **[ ]** Grouped by 6s: 3 builders for every 2 speedrunners (5 total per group).	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**18 [ENTER] 12 [g] [a:b]**): Shows **3:2** (Part-to-Part). For Part-to-Whole (**18 [ENTER] 30 [g] [a:b]**): shows **3:5**.	3. [PROVE] VERIFIED SOLUTION: 3 + 2 = 5 total parts!

Problem 2 (Tape Diagram without Drawing): The ratio of wins to losses for a basketball team is 5:3. If the team played 32 games total, how many games did they win?

1. [BLOCKS] TOUCH IT: **[ ]** 5 Blue tiles + 3 Gold tiles = 8 total tiles.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**32 [ENTER] 5 [ENTER] 3 [+] [÷] 5 [×]**): Snapshot per tile: **4 games/tile**. Multiply by 5 win tiles -> **20 wins** (and 3 × 4 = 12 losses).	3. [PROVE] VERIFIED SOLUTION: Simplifies back to **5:3** (and 20 + 12 = 32)!

Problem 3 (Boss Level — Difference Between Parts): The ratio of acoustic guitars to electric guitars in a shop is 2:5. There are 18 more electric guitars than acoustic guitars. How many electric guitars are there?

1. [BLOCKS] TOUCH IT: 5 Gold tiles minus 2 Blue tiles = 3 extra tiles. Those 3 extra tiles equal 18 guitars!	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**18 [ENTER] 5 [ENTER] 2 [-] [÷] 5 [×]**): **18 ÷ 3 = 6** guitars per tile -> 6 × 5 = **30 electric guitars** (and 12 acoustic). • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[BUILDER]** Tile Gap Model): 7 parts - 4 parts = 3 extra parts = 18 -> 1 part = 6 -> 11 parts = 66. Way B (**[STACK-HACK]** Stack Ratio): **18 [ENTER] 3 [÷] 11 [×]** = 66 guitars!	3. [PROVE] VERIFIED SOLUTION: 30 - 12 = 18, and **12 [ENTER] 30 [g] [a:b]** -> **2:5**!

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[BUILDER]** Tile Gap Model): 7 parts - 4 parts = 3 extra parts = 18 -> 1 part = 6 -> 11 parts = 66. Way B (**[STACK-HACK]** Stack Ratio): **18 [ENTER] 3 [÷] 11 [×]** = 66 guitars!

Day 12 (Lesson 3.2): Ratio Tables & Double Number Lines via Rhythmic Skip-Counting ([g] [REP])

SPIRAL PATH: Thread C (Scale & Ratio) • [[SPARK] VISIT 2: PATTERN WAY — Revisits Day 2 REP & Day 11 Tape Diagrams] // VARIETY MODE: [BOOMERANG] Mode D: 5-Second Ratio Table Sprint ({{key:REP}})

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Type **4 [g] [REP]** (**[g][ENTER]**) and tap **[+]** four times: watch **8 -> 12 -> 16 -> 20** march across the screen! You just built a 5-column ratio row in 3 seconds! (**[ ] +5 Spark XP!**)

The Real Question: Why does the textbook make you fill out Ratio Tables AND Double Number Lines—and how does the SC-6 **[g] [REP]** key generate every column in rhythm?

Why Should I Care? (2-Min Hook): A Ratio Table and a Double Number Line are the exact same thing: one is written in boxes, and the other has arrows on a line! Both are just two skip-counting ladders running side by side. And the SC-6 has a dedicated Skip-Counting Ladder key (**[g] [REP]**, **[g][x⊠y]**) that steps through multiples with every click of **[+]**!
Why Try a New Way Today? (When the Old Way Gets Annoying): On Day 11, laying out tiles (**[BUILDER]**) worked great for small numbers. But if a ratio table jumps to 10 or 15 batches, you don't want to count 90 plastic tiles! Using Day 2's **[g] [REP]** constant key (**[SPARK]**) lets you skip-count an entire ratio table in 5 seconds!

Batch #	×1	×2	×3	×4	×5	How to Generate on SC-6 ([g] [REP])
Citric Acid (Top Line)	**3**	**6**	**9**	**12**	**15**	Key **3 [g] [REP]**, then tap **[+]** -> 6 -> 9 -> 12 -> 15!
Baking Soda (Bottom Line)	**7**	**14**	**21**	**28**	**35**	Key **7 [g] [REP]**, then tap **[+]** -> 14 -> 21 -> 28 -> 35!

Problem 1 (Dual Skip-Count Ladder): A drone flies 8 meters every 3 seconds. Use [g] [REP] to find how far it flies in 15 seconds.

1. [BLOCKS] TOUCH IT: **[ ]** 5 jumps of 3 seconds (3, 6, 9, 12, 15).	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**8 [g] [REP] [+] [+] [+] [+]**): Steps **8 -> 16 -> 24 -> 32 -> 40**.	3. [PROVE] VERIFIED SOLUTION: Simplifies right back to **8:3**. Final Answer: **40 meters**.

Problem 2 (Missing Box in a Ratio Table): In a ratio table where A : B = 4 : 9, find the missing value when A = 28.

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**28 [ENTER] 4 [÷] 9 [×]**): Scale multiplier is **7** (28 ÷ 4 = 7). Snapshot **Y: 7 | X: 9** -> **63**.	3. [PROVE] VERIFIED SOLUTION: Screen confirms **4:9**!

Problem 3 (Boss Level — Scaling Down then Up): 6 tickets to a skate park cost $45. How much do 10 tickets cost? (Hint: Divide by 2 to get 3 tickets, or divide by 6 to get 1 ticket, then multiply by 10!)

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**45 [ENTER] 6 [÷] 10 [×]**): Snapshot **Y: 7.50 | X: 10** -> **75.00**. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[SPARK]** Scale Down to 2 Tickets): 6 tickets = 45 -> 2 tickets = 15 -> 10 tickets (5 × 2) = 75. Way B (**[STACK-HACK]** Unit Rate Stack): **45 [ENTER] 6 [÷] 10 [×]** = $75!	3. [PROVE] VERIFIED SOLUTION: Returns **45**!

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[SPARK]** Scale Down to 2 Tickets): 6 tickets = 45 -> 2 tickets = 15 -> 10 tickets (5 × 2) = 75. Way B (**[STACK-HACK]** Unit Rate Stack): **45 [ENTER] 6 [÷] 10 [×]** = 75!

Day 13 (Lesson 3.3): Unit Rates & Best-Buy Showdowns (Plus the [x⇄y] Swap Rescue!)

SPIRAL PATH: Thread C & B Boomerang • [[STACK-HACK] VISIT 3: STACK HACKER WAY — Revisits Days 8, 11 & 12] // VARIETY MODE: [DETECTIVE] Mode A: Best-Buy Store Showdown ({{key:x⇄y}} Rescue)

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Oops—you typed **5 [ENTER] 15** instead of **15 [ENTER] 5**! Don't clear it—just press **[x⇄y]** (**[f][R↓]**) to swap Floors 1 & 2, then press **[÷]** -> **3**! (**[ ] +5 Spark XP!**)

The Real Question: When comparing two store deals, how do you know which number goes in first for "Dollars PER Ounce"—and how does **[x⇄y]** save you if you typed them backwards?

Why Should I Care? (2-Min Hook): Stores try to trick you with weird package sizes: a 12 oz bag of chips for $4.20 vs. a 16 oz bag for $5.12. Which one is actually cheaper per ounce? And if you accidentally type ounces first and dollars second, you don't have to clear and retype—just hit the Swap key **[x⇄y]** to flip Floor 1 and Floor 2 before hitting **[÷]**!
Why Try a New Way Today? (When the Old Way Gets Annoying): Tape diagrams (**Day 11**) and skip-counting tables (**Day 12**) get stuck when comparing weird pack sizes like **8 oz for 3.60** vs **12 oz for 5.04**. Dividing down to 1 single unit (**[STACK-HACK]** Unit Rate) works on any two numbers—and **[x⇄y]** rescues you if you enter them upside-down!

Store Option	Keystroke Sequence	Unit Price ( )	Winner
Bag A ($4.20 for 12 oz)	**4.20 [ENTER] 12 [÷]**	**$0.35** / oz	Good
Bag B ($5.12 for 16 oz)	**5.12 [ENTER] 16 [÷]** (keep **0.35** on **Y**!)	**$0.32** / oz	BEST BUY! (Saves $0.03/oz)

Problem 1 (The [x⇄y] Swap Rescue): You want the unit rate in words per minute for typing 315 words in 5 minutes. Suppose you accidentally typed 5 [ENTER] 315. Fix it with one key!

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**5 [ENTER] 315 [x⇄y]**): Snapshot after **[x⇄y]**: **Y: 315 | X: 5**. Now press **[÷]** -> **63 words/min**.	3. [PROVE] VERIFIED SOLUTION: 63 × 5 = 315!

Problem 2 (Best-Buy Showdown): Pack A has 8 batteries for $11.60. Pack B has 12 batteries for $16.80. Find both unit prices on the stack at the same time and name the better buy.

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**11.60 [ENTER] 8 [÷] 16.80 [ENTER] 12 [÷]**): Snapshot **Y (Pack A): 1.45 | X (Pack B): 1.40**.	3. [PROVE] VERIFIED SOLUTION: Pack B (**1.40/battery**) beats Pack A (**1.45/battery**) by **$0.05** per battery!

Problem 3 (Boss Level — Rate Prediction): A 3D printer extrudes 45 millimeters of filament in 15 seconds. At this rate, how many millimeters will it extrude in 120 seconds (2 minutes)?

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**45 [ENTER] 15 [÷] 120 [×]**): Snapshot **Y (Unit Rate): 3 mm/s | X: 120** -> **360 mm**. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[STACK-HACK]** Unit Rate **mm/s**): **45 [ENTER] 15 [÷]** = 3 mm/s × 60 s = 180 mm. Way B (**[SPARK]** Ratio Table Scaling from Day 12): 60 s is 4 × 15 s, so 4 × 45 mm = 180 mm!	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[STACK-HACK]** Unit Rate **mm/s**): **45 [ENTER] 15 [÷]** = 3 mm/s × 60 s = 180 mm. Way B (**[SPARK]** Ratio Table Scaling from Day 12): 60 s is 4 × 15 s, so 4 × 45 mm = 180 mm!

Day 14 (Lesson 3.4): Percents, Fractions & Decimals — The 3-Way Costume Change

SPIRAL PATH: Thread C (Scale & Ratio) • [[BOOMERANG] VISIT 4: CROSS-CHAPTER BOOMERANG — Revisits Day 10 FDISP & Day 11 Ratios] // VARIETY MODE: [SPARK] Mode C: 3-Costume Instant Reveal ({{key:%}})

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Hold up the **1/4** Fraction Tower. You know 1 quarter is 25 cents (25\%). Now type **[.] 1 [.] 4 [ENTER] [f] [%]** (**[f][%]**)—screen says **25%**! (**[ ] +5 Spark XP!**)

The Real Question: Why are 3/4, 0.75, and 75\% literally the exact same battery charge—and how do you switch between all three on the SC-6?

Why Should I Care? (2-Min Hook): Think of your phone or tablet battery bar: when it's 3/4 full, the corner says **75%**, which as a decimal is **0.75**. "Per-cent" literally means "out of 100" (like 100 cents in a dollar!). They aren't three different math topics—they are one single amount wearing three different outfits!
Why Try a New Way Today? (When the Old Way Gets Annoying): Percents aren't a brand-new topic—they are just Day 11 ratios locked to a denominator of **100**! Instead of memorizing 6 rules for moving decimal points left or right, use **[f] [%]** and **[FDISP]** (**[STACK-HACK]**) to see all 3 costumes of the same number!

Fraction Tower ([FRAC-01])	Fraction ([FDISP])	Decimal ([NUM])	Percent ([f] [%])	Money Analogy ([$])
Half Tower (42.0 mm)	**1/2**	**0.50**	**50%**	2 quarters (**$0.50**)
Quarter Tower (21.0 mm)	**1/4**	**0.25**	**25%**	1 quarter (**$0.25**)
Three-Quarters (63.0 mm)	**3/4**	**0.75**	**75%**	3 quarters (**$0.75**)
Fifth equivalences	**3/5**	**0.60**	**60%**	6 dimes (**$0.60**)

Problem 1 (The 3-Outfit Table): Convert 7/20 into a decimal and a percent.

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**[.] 7 [.] 20 [ENTER] [FDISP]**): Decimal shows **0.35**. Press **[f] [%]** (**[f][%]**) -> **35%**.	3. [PROVE] VERIFIED SOLUTION: Turns **35%** into **0.35** and back into **7/20**!

Problem 2 (Free-Throw Percentage): You made 18 out of 24 free throws. What is your simplified fraction, decimal, and shooting percentage?

1. [BLOCKS] TOUCH IT: **[ ]** 18/24 reduces to three 1/4 blocks (3/4).	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**18 [ENTER] 24 [÷] [f] [%]**): Decimal: **0.75** (**3/4**). Percent: **75%**.	3. [PROVE] VERIFIED SOLUTION: 75\% of 24 = 18 shots made!

Problem 3 (Boss Level — Ordering Mixed Formats): Put 5/8, 60\%, and 0.61 in order from least to greatest by converting all three to decimals on the stack.

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: **[.] 5 [.] 8 [ENTER] [FDISP]** -> **0.625**; **60 [%]** -> **0.60**; **0.61**. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[SPARK]** All-Decimals Lineup): 60\% = 0.60, 0.61, 5/8 = 0.625. Way B (**[STACK-HACK]** All-Percents via **[f] [%]**): 60\% < 61\% < 62.5\%. Both prove **60%, 0.61, 5/8**!	3. [PROVE] VERIFIED SOLUTION: **60% (0.60) < 0.61 < 5/8 (0.625)**.

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[SPARK]** All-Decimals Lineup): 60\% = 0.60, 0.61, 5/8 = 0.625. Way B (**[STACK-HACK]** All-Percents via **[f] [%]**): 60\% < 61\% < 62.5\%. Both prove **60%, 0.61, 5/8**!

Day 15 (Lesson 3.5): Discounts, Tips, Percent Change & Unit Conversions

SPIRAL PATH: Thread C (Scale & Ratio) • [[STACK-HACK] VISIT 5: BOSS BOOMERANG — Revisits Days 11–14] // VARIETY MODE: [BOOMERANG] Mode D: 2-Key Sale Price Hack ({{key:%}} & {{key:%CHG}})

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): A $50 hoodie is 20\% off. Type **50 [ENTER] 20 [%]**—Floor 1 shows the **10** discount while **50** waits on Floor 2! Press **[-]** -> **$40** sale price! (**[ ] +5 Spark XP!**)

The Real Question: How do you find a sale price, calculate percent change (**[g] [%CHG]**), or convert inches to centimeters (**[f] [⌧cm]**) in two button presses?

Why Should I Care? (2-Min Hook): Whether you're buying a $60 video game at 25\% off or converting a 3D print from inches to millimeters/centimeters, this is the math you actually use every week of your life.
Why Try a New Way Today? (When the Old Way Gets Annoying): Finding a sale price usually takes two separate steps (find discount, then subtract). Today you compare two pro ways: subtracting the **[%]** discount on **Y | X** (**[STACK-HACK]**) vs multiplying by the remaining percent (**[SPARK]**), plus 1-click unit conversions (**[f] [⌧cm]**, **[f] [⌧kg]**)!

Step	Keystroke	Floor 2 ( )	Floor 1 ( )	What Just Happened?
1	**60 [ENTER]**	**60**	**60**	Original price $60 sits on Floor 2 (**Y**).
2	**25 [%]**	**60**	**15**	Magic! **X** shows the $15 discount while **Y** still holds **$60**!
3	**[-]**	**0**	**45**	Subtracts $60 - 15 = 45 Sale Price!

Problem 1 (Sneaker Sale): A $80 pair of skate shoes is on sale for 30\% off. Find (a) the discount amount and (b) the final sale price.

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**80 [ENTER] 30 [%]**): Snapshot before **[-]**: **Y (Original): 80 | X (Discount): 24**. Press **[-]** -> **$56.00**.	3. [PROVE] VERIFIED SOLUTION: Screen displays **-30%** (a 30\% decrease)!

Problem 2 (Percent Change Trophy): A YouTube channel grew from 250 subscribers to 325 subscribers. What was the percent increase?

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**250 [ENTER] 325 [g] [%CHG]**): Snapshot **Y: 250 | X: 325** -> **+30%**.	3. [PROVE] VERIFIED SOLUTION: 250 + 75 = 325!

Problem 3 (Boss Level — Metric/Customary Lab Conversion): A skateboard deck is 32 inches long and weighs 5 pounds. Convert its length to centimeters ([f] [⌧cm]) and its weight to kilograms ([f] [⌧kg]).

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: **32 [f] [⌧cm]** -> **81.28 cm**. **5 [f] [⌧kg]** -> **2.27 kg**. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[STACK-HACK]** 1-Click Converter Keys): **32 [f] [⌧cm]** = **81.28 cm** and **5 [f] [⌧kg]** = **2.27 kg**. Way B (**[SPARK]** Unit-Rate Multiply from Day 13): **32 [ENTER] 2.54 [×]** = **81.28 cm**!	3. [PROVE] VERIFIED SOLUTION: Press **[g] [⌧in]** on **81.28** -> returns to **32**! Press **[g] [⌧lb]** on **2.268** -> returns to **5**!

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[STACK-HACK]** 1-Click Converter Keys): **32 [f] [⌧cm]** = **81.28 cm** and **5 [f] [⌧kg]** = **2.27 kg**. Way B (**[SPARK]** Unit-Rate Multiply from Day 13): **32 [ENTER] 2.54 [×]** = **81.28 cm**!

CHAPTER 4: Algebraic Expressions & Properties (Days 16–19)

Aligned to Big Ideas Math Grade 6, Chapter 3 / Chapter 5: Writing & Evaluating Algebraic Expressions, Combining Like Terms, The Distributive Property, and Factoring Expressions.

Day 16 (Lesson 4.1): Evaluating Algebraic Expressions — Plugging Numbers Into the Stack Elevator

SPIRAL PATH: Thread D (Inside-Out Chunking) • [[SPARK] VISIT 2: PATTERN WAY — Revisits Day 3 4-Stack Order of Operations] // VARIETY MODE: DICE Mode E: Secret Variable Box Drop (ALGB-01)

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Drop 4 unit cubes into the green **x** box. Now 'read' 2x + 3: two boxes (8) plus 3 = **11**! Key **2 [ENTER] 4 [×] 3 [+]** -> **11**! Algebra is just a box with a number inside! (**[ ] +5 Spark XP!**)

The Real Question: What is a variable like x really doing—and how do you plug a number into an expression like 3x^2 - 5y without writing 4 lines of substitution steps?

Why Should I Care? (2-Min Hook): Kids get scared the moment letters show up in math. Here is the secret: a letter like x is just an empty slot on the SC-6 elevator waiting for you to drop a number block into it! And when you see **3x** stuck together with no symbol between them, it just means **3 × x**!
Why Try a New Way Today? (When the Old Way Gets Annoying): Algebra letters (**x**, **y**) look scary until you realize a variable is just an empty green box (**ALGB-01**) waiting for a number! Once you drop the number into the box (**[BUILDER]**), evaluating 3x^2 - 5 is literally just Day 3's elevator math (**[STACK-HACK]**)!

Camera Angle	What You Do	What You See
Angle 1: Physical Algebra Tiles (**ALGB-01**)	Lay out four green x-rods and seven yellow 1-unit cubes. Now replace each x-rod with a row of **5** units!	Four rows of 5 (= 20) plus 7 units = 27!
Angle 2: Textbook Substitution	Replace x with (5): 4(5) + 7 = 20 + 7 = 27.	Parentheses around **(5)** mean multiply!
Angle 3: SC-6 Stack Evaluation	Type **4 [ENTER] 5 [×]** (-> 20 on Stack), then **7 [+]** -> **27**!	Snapshot before final **[+]**: **Y: 20 | X: 7**.

Problem 1 (Single Variable Plug-In): Evaluate 6x - 9 when x = 7.

1. [BLOCKS] TOUCH IT: **[ ]** Six x-rods (each worth 7) minus 9 units.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**6 [ENTER] 7 [×] 9**): Snapshot before **[-]**: **Y: 42 | X: 9**. Press **[-]** -> **33**.	3. [PROVE] VERIFIED SOLUTION: Restores **42**, then divides by **7** to restore coefficient **6**!

Problem 2 (Two-Variable Expression): Evaluate 3a^2 + 4b when a = 5 and b = 6. (Watch out: in 3a^2, square a FIRST, then multiply by 3!)

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**5 [x²] 3 [×] 4 [ENTER] 6 [×]**): Snapshot before **[+]**: **Y (3a²): 75 | X (4b): 24**. Press **[+]** -> **99**.	3. [PROVE] VERIFIED SOLUTION: Subtracts **24** and restores **75**!

Problem 3 (Boss Level — Skate Park Rental Formula): Renting a helmet costs $8 flat plus $4.50 per hour (8 + 4.50h). How much does a 3.5-hour session cost?

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**8 [ENTER] 4.50 [ENTER] 3.5 [×]**): Snapshot before **[+]**: **Y: 8.00 | X: 15.75**. Press **[+]** -> **23.75**. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[STACK-HACK]** Inside-Out Formula Stack): **4.5 [ENTER] 4 [×] 8 [+]** = 26. Way B (**[BUILDER]** Repeat Addition Table from Day 12): $8 + 4.50 + 4.50 + 4.50 + 4.50 = 8 + 18 = $26!	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[STACK-HACK]** Inside-Out Formula Stack): **4.5 [ENTER] 4 [×] 8 [+]** = 26. Way B (**[BUILDER]** Repeat Addition Table from Day 12): 8 + 4.50 + 4.50 + 4.50 + 4.50 = 8 + 18 = $26!

Day 17 (Lesson 4.2): Combining Like Terms — Why x^2 Flats, x Rods, and 1 Cubes Never Mix

SPIRAL PATH: Thread D (Inside-Out Chunking) • [[BUILDER] VISIT 1: BUILDER WAY — Tactile Like-Term Sorting] // VARIETY MODE: [DETECTIVE] Mode A: Spot the Glitch ('Does 3x + 2 = 5x?')

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Put 3 green x-rods and 2 yellow unit cubes on your desk. Are there 5 green rods? NO! Now test x=10 on your SC-6: 3(10)+2 = 32, while 5(10) = 50! Different shapes never merge! (**[ ] +5 Spark XP!**)

The Real Question: Why does 3x + 2x = 5x, while 3x + 2 CANNOT be combined into 5x—and why can't x^2 and x combine?

Why Should I Care? (2-Min Hook): The biggest mistake in middle school algebra is writing 3x + 2 = 5x or 2x^2 + 4x = 6x^3. When you open the Algebra Boxes & Tiles (**ALGB-01**), you see why immediately:
Why Try a New Way Today? (When the Old Way Gets Annoying): On paper, kids constantly add 3x + 5 and write 8x. Once you hold a green x-rod (42 mm) and a small yellow 1-cube (14 mm) (**[BUILDER]**), your hands see why they can't merge—and plugging a test number into the SC-6 (**[STACK-HACK]**) proves your simplified expression matches!

Camera Angle	What You Do	What You See
Angle 1: Physical Tile Sorting (**ALGB-01**)	Put **4** x-rods and **5** unit cubes on the left; **2** x-rods and **3** unit cubes on the right. Now slide all the rods into Pile A and all the cubes into Pile B!	Pile A has 6 x-rods (6x). Pile B has 8 unit cubes (+8). Total: 6x + 8!
Angle 2: SC-6 Two-Floor Ledger	Add the x-coefficients (**4 [ENTER] 2 [+]** -> 6). Then add the constant cubes (**5 [ENTER] 3 [+]** -> 8).	Look at your screen: Floor 2 (**Y**) holds **6** (the x-rods) and Floor 1 (**X**) holds **8** (the constant units)—sitting on separate floors so you never mash them together!
Angle 3: The Test-Number Proof (x = 10)	Plug x = 10 into the original expression vs. your simplified 6x + 8!	Original: 4(10)+5+2(10)+3 = 40+5+20+3 = 68. Simplified: 6(10)+8 = 68. Identical!

Problem 1 (Rods & Cubes Sort): Simplify 7x + 4 - 2x + 9.

1. [BLOCKS] TOUCH IT: **[ ]** Sorted x-rods (7 - 2) and unit cubes (4 + 9).	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**7 [ENTER] 2 [-] 4 [ENTER] 9 [+]**): Snapshot **Y (x-rods): 5 | X (units): 13** -> **5x + 13**.	3. [PROVE] VERIFIED SOLUTION: 5(10) + 13 = 63, and 70 + 4 - 20 + 9 = 63!

Problem 2 (Squares, Rods & Cubes): Simplify 3x^2 + 5x + 2x^2 - x + 6.

1. [BLOCKS] TOUCH IT: **[ ]** Three piles: x^2-flats (3+2=5), x-rods (5-1=4), 1-cubes (6).	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: **Z (x²): 5 | Y (x): 4 | X (units): 6** -> **5x² + 4x + 6**.	3. [PROVE] VERIFIED SOLUTION: 5(4) + 4(2) + 6 = 34, matching 3(4)+5(2)+2(4)-2+6 = 34!

Problem 3 (Boss Level — Perimeter of a Tile Rectangle): A rectangle has two sides of length (2x + 3) and two sides of length (x + 5). Write the simplified perimeter expression.

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: x-rods: **2 [ENTER] 2 [+] 1 [+] 1 [+]** (-> 6). Units: **3 [ENTER] 3 [+] 5 [+] 5 [+]** (-> 16). Snapshot **Y: 6 | X: 16** -> **6x + 16**. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[BUILDER]** Sort All 4 Sides by Tile Shape): (2x+2x+x+x) + (3+3+5+5) = **6x + 16**. Way B (**[STACK-HACK]** Numerical Twin Proof at x=10): Original 2(23) + 2(15) = 76 == Simplified 6(10) + 16 = 76!	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[BUILDER]** Sort All 4 Sides by Tile Shape): (2x+2x+x+x) + (3+3+5+5) = **6x + 16**. Way B (**[STACK-HACK]** Numerical Twin Proof at x=10): Original 2(23) + 2(15) = 76 == Simplified 6(10) + 16 = 76!

Day 18 (Lesson 4.3): The Distributive Property — One Split Rectangle, Two Equivalent Ways

SPIRAL PATH: Thread D (Inside-Out Chunking) • [[SPARK] VISIT 2: PATTERN WAY — Revisits Day 3 Grouping & Day 17 Tiles] // VARIETY MODE: [SPARK] Mode C: Split-Rectangle Twin Match (a(b+c))

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Build 2 rows of (1 green x-rod + 3 yellow cubes). Count by rows: 2(x+3). Count by color piles: **2x + 6**! Same blocks on your desk, two ways to name them! (**[ ] +5 Spark XP!**)

The Real Question: Why does the textbook force you to calculate 6(10 + 4) two different ways—and how does one split rectangle of tiles prove they are identical?

Why Should I Care? (2-Min Hook): Few things annoy middle schoolers more than a problem that says: "Use the Distributive Property to find 6 × 14." The kid thinks: "Why can't I just multiply 6 × 14 = 84?!" Here is why: when an expression has a variable inside like 6(x + 4), you can't add x + 4 first! The 4-Color Array Tiles (**COUNT-01**) and Algebra Stage (**ALGB-01**) show that multiplying the whole width at once or multiplying the two split rooms separately builds the exact same rectangle!
Why Try a New Way Today? (When the Old Way Gets Annoying): Why does 3(x + 4) = 3x + 12? Because whether you measure the whole rectangle at once (**Way A**) or split it into two rooms (**Way B**), the total tile count never changes! This is the #1 place in middle school where two approaches prove each other.

Camera Angle	What You Do	What You See
Angle 1: Physical Algebra Stage (**ALGB-01**)	Build 3 identical rows. In each row, place one x-rod and four 1-unit cubes.	Look at it by rows: 3 rows of (x + 4) -> 3(x + 4). Look at it by columns: 3 x-rods plus 12 unit cubes (3 × 4) -> 3x + 12!
Angle 2: Mental Math Split (6 × 14)	Split **14** into **10 + 4**. Multiply 6 × 10 = 60 and 6 × 4 = 24, then add 60 + 24 = 84.	On SC-6: **6 [ENTER] 10 [×] 6 [ENTER] 4 [×]** leaves **Y: 60 | X: 24**. Press **[+]** -> **84**!
Angle 3: Direct Check	Compare against **6 [ENTER] 10 [ENTER] 4 [+] [×]**.	Snapshot **Y: 6 | X: 14** -> **84**. Both routes land on the exact same number!

Problem 1 (Two Ways on the Stack): Compute 7 × (20 + 3) both ways on your SC-6 and record both snapshots.

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Way A (Add Inside First): **7 [ENTER] 20 [ENTER] 3 [+]** -> Snapshot **Y: 7 | X: 23** -> **[×]** = **161**. • [STACK] Way B (Distribute to Both Rooms): **7 [ENTER] 20 [×] 7 [ENTER] 3 [×]** -> Snapshot **Y: 140 | X: 21** -> **[+]** = **161**.	3. [PROVE] VERIFIED SOLUTION: Both snapshots produce **161**!

Problem 2 (Algebra Tile Distribution): Build 4 rows of (x + 5) on the Algebra Stage and write the distributed expression for 4(x + 5).

1. [BLOCKS] TOUCH IT: **[ ]** 4 rows × 1 rod = 4x; 4 rows × 5 cubes = 20.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**4 [ENTER] 1 [×] 4 [ENTER] 5 [×]**): Snapshot **Y (x-coeff): 4 | X (constant): 20** -> **4x + 20**.	3. [PROVE] VERIFIED SOLUTION: 4(3 + 5) = 4(8) = 32, and 4(3) + 20 = 12 + 20 = 32!

Problem 3 (Boss Level — Distribute & Combine): Simplify 5(2x + 3) + 4x.

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: Distribute 5: **5 [ENTER] 2 [×]** (-> 10x) and **5 [ENTER] 3 [×]** (-> 15). Add **4x** to **10x**: **10 [ENTER] 4 [+]** -> 14. Snapshot **Y: 14 | X: 15** -> **14x + 15**. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[BUILDER]** Distribute & Sort Tiles): 5(2x+3) + 4x = 10x + 15 + 4x = **14x + 15**. Way B (**[STACK-HACK]** Numerical Twin Proof at x=2): Original 5(7) + 8 = 43 == Simplified 14(2) + 15 = 43!	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[BUILDER]** Distribute & Sort Tiles): 5(2x+3) + 4x = 10x + 15 + 4x = **14x + 15**. Way B (**[STACK-HACK]** Numerical Twin Proof at x=2): Original 5(7) + 8 = 43 == Simplified 14(2) + 15 = 43!

Day 19 (Lesson 4.4): Factoring Expressions with GCF — Running the Distributive Property in Reverse

SPIRAL PATH: Thread A & D Boomerang • [[BOOMERANG] VISIT 4: CROSS-CHAPTER BOOMERANG — Revisits Day 5 GCF & Day 18 Distributive] // VARIETY MODE: [TARGET] Mode B: Reverse the Rectangle ({{key:GCF}} Unlock)

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Take 4 green x-rods and 6 yellow cubes (4x + 6). Arrange them into 2 identical rows—each row has **2x + 3**! Now press **4 [ENTER] 6 [GCF]** -> **2** (the number of rows)! (**[ ] +5 Spark XP!**)

The Real Question: How do you factor an expression like 18x + 24 into 6(3x + 4) without guessing—using the **[GCF]** key?

Why Should I Care? (2-Min Hook): "Factoring" sounds scary, but it is literally just putting the tiles back into equal rows! Yesterday you multiplied the row count into the parentheses: 6(3x + 4) -> 18x + 24. Today, you use the SC-6 **[GCF]** button to find the biggest number of equal rows you can stack 18x + 24 into!
Why Try a New Way Today? (When the Old Way Gets Annoying): Remember **[GCF]** from Chapter 1 (**Day 5**)? Today it boomerangs back to unlock Algebra! Factoring 45x - 30 isn't a brand-new process—it's just running Day 18's Distributive Property in reverse using Day 5's **[GCF]** key (**[STACK-HACK]**)!

Problem 1 (Whole-Number GCF Pull): Factor 12 + 20 into the form a(b + c) using [GCF].

1. [BLOCKS] TOUCH IT: **[ ]** Arranged 12 Blue tiles and 20 Gold tiles into 4 equal rows of (3 Blue + 5 Gold).	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**12 [ENTER] 20 [GCF]**): GCF is **4**. Inside parts: 12 ÷ 4 = 3 and 20 ÷ 4 = 5 (**Y: 3 | X: 5**). Result: **4(3 + 5)**.	3. [PROVE] VERIFIED SOLUTION: 4 × 8 = 32, and 12 + 20 = 32!

Problem 2 (Algebraic Factoring): Factor 24x + 36 completely using [GCF].

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**24 [ENTER] 36 [GCF]**): GCF is **12**. Divide both by 12 (**24 ÷ 12 = 2**, **36 ÷ 12 = 3**). Snapshot **Z (GCF): 12 | Y: 2 | X: 3** -> **12(2x + 3)**.	3. [PROVE] VERIFIED SOLUTION: Distribute 12 × 2 = 24x and 12 × 3 = 36!

Problem 3 (Boss Level — Subtraction Factoring): Factor 45x - 30 completely.

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**45 [ENTER] 30 [GCF]**): GCF is **15**. Inside: 45 ÷ 15 = 3x and 30 ÷ 15 = 2. Final Answer: **15(3x - 2)**. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[STACK-HACK]** SC-6 **[GCF]** Pull-Out): **45 [ENTER] 30 [GCF]** = **15** -> **15(3x - 2)**. Way B (**[SPARK]** Multiply Back Out from Day 18): 15 × 3x = 45x and 15 × 2 = 30 -> **45x - 30**!	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[STACK-HACK]** SC-6 **[GCF]** Pull-Out): **45 [ENTER] 30 [GCF]** = **15** -> **15(3x - 2)**. Way B (**[SPARK]** Multiply Back Out from Day 18): 15 × 3x = 45x and 15 × 2 = 30 -> **45x - 30**!

CHAPTER 5: Areas of Polygons & Elapsed Time (Days 20–23)

Aligned to Big Ideas Math Grade 6, Chapter 4: Areas of Parallelograms, Triangles, Trapezoids, Composite Figures, Coordinate Polygons & Base-60 Time.

Day 20 (Lesson 5.1): Areas of Parallelograms & Rectangles — Sliding the Corner Triangle (`[f] [RECT]`)

SPIRAL PATH: Thread D (Inside-Out Chunking) • [[BUILDER] VISIT 1: BUILDER WAY — Geometric Area Arrays] // VARIETY MODE: [DETECTIVE] Mode A: Slant-Side Trap Detector ({{key:RECT}})

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Snap a 5 by 3 rectangle of tiles (15 tiles). Press **`5 [ENTER] 3 [f] [RECT]`** (**`[f][1]`**) -> **`15`**! Even if you slant the stack sideways into a parallelogram, did the number of plastic tiles change? Never! (**`[ ] +5 Spark XP!`**)

The Real Question: Why does a slanted parallelogram have the exact same area formula (A = b × h) as a straight rectangle—and why do you ignore the slanted side length?

Why Should I Care? (2-Min Hook): Textbooks always put a trap number on parallelogram problems: they label the base (10 cm), the straight vertical height (6 cm), AND the slanted side (7 cm). Kids who guess which numbers to multiply pick the slanted 7 cm and get it wrong! With physical tiles, you see that chopping the triangle off one slanted end and sliding it to the other end turns every parallelogram into a straight Rectangle (**`[f] [RECT]`**)!
Why Try a New Way Today? (When the Old Way Gets Annoying): Kids accidentally multiply the slanted side of a parallelogram instead of the straight height. Sliding the corner triangle on the grid (**`[BUILDER]`**) proves every parallelogram is a disguised rectangle—so **`[f] [RECT]`** (**`[STACK-HACK]`**) solves both!

Camera Angle	What You Do	What You See
Angle 1: Physical Tile Shear (**`COUNT-01`**)	Build a 12 × 8 rectangle of tiles (96 tiles). Push the rows slightly sideways so the sides slant.	Did any tiles fall off the desk? No! There are still 96 tiles! The straight height is still 8 rows. Ignore the slanted edge!
Angle 2: SC-6 Dedicated **`[f] [RECT]`** Key	Key **`12 [ENTER] 8 [f] [RECT]`** (**`[f][7]`**).	**`[f] [RECT]`** multiplies Base (**`Y`**) by Height (**`X`**) -> **`96`**.
Angle 3: Finding a Missing Height	If Area = 96 and Base = 12, press **`[f] [LASTX]`** or **`96 [ENTER] 12 [÷]`** -> **`8`**.	Division undoes area immediately!

Problem 1 (Dodge the Slanted Trap!): A parallelogram has base b = 15 m, vertical height h = 9 m, and slanted side 11 m. Find the area.

1. [BLOCKS] TOUCH IT: **`[ ]`** Crossed out the trap number **`11 m`** (slanted edge); kept **`15`** and **`9`**.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**`15 [ENTER] 9 [f] [RECT]`**): Snapshot **`Y (Base): 15 | X (Height): 9`** -> **`135 m²`**.	3. [PROVE] VERIFIED SOLUTION: 135 ÷ 9 = 15!

Problem 2 (Decimal Courtyard): A rectangular maker lab measures 8.5 m by 6.4 m. Find its floor area.

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**`8.5 [ENTER] 6.4 [f] [RECT]`**): Snapshot **`Y: 8.5 | X: 6.4`** -> **`54.4 m²`**.	3. [PROVE] VERIFIED SOLUTION: Restores **`8.5`**!

Problem 3 (Boss Level — Missing Base): A parallelogram banner has an area of 168 in^2 and a vertical height of 14 in. How long is its base?

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**`168 [ENTER] 14 [÷]`**): Snapshot **`Y: 168 | X: 14`** -> **`12 inches`**. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**`[STACK-HACK]`** Inverse Division): **`168 [ENTER] 12 [÷]`** = **`14 in`**. Way B (**`[SPARK]`** Forward **`[f] [RECT]`** Proof): **`14 [ENTER] 12 [f] [RECT]`** = **`168 in²`**!	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**`[STACK-HACK]`** Inverse Division): **`168 [ENTER] 12 [÷]`** = **`14 in`**. Way B (**`[SPARK]`** Forward **`[f] [RECT]`** Proof): **`14 [ENTER] 12 [f] [RECT]`** = **`168 in²`**!

Day 21 (Lesson 5.2): Areas of Triangles — Why Every Triangle Is Half a Rectangle ([g] [TRI])

SPIRAL PATH: Thread D (Inside-Out Chunking) • [[SPARK] VISIT 2: PATTERN WAY — Revisits Day 20 Parallelogram/Rectangle] // VARIETY MODE: [BOOMERANG] Mode D: Half-a-Rectangle Shortcut ({{key:TRI}})

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): A 6 by 4 rectangle has area **24** (**6 [ENTER] 4 [f] [RECT]**). Cut it corner-to-corner into two identical triangles—each triangle is half of 24 = **12**! Now press **6 [ENTER] 4 [g] [TRI]** (**[g][1]**) -> **12**! (**[ ] +5 Spark XP!**)

The Real Question: Why do you always divide by 2 for a triangle's area—and how does the SC-6 **[g] [TRI]** key (**[g][7]**) keep you from ever forgetting the 1/2?

Why Should I Care? (2-Min Hook): The #1 mistake on 6th-grade geometry tests is multiplying b × h for a triangle and forgetting to divide by 2! Look at your SC-6 keypad on key **7**:
Why Try a New Way Today? (When the Old Way Gets Annoying): Why does a triangle have a ÷ 2 in its formula when a rectangle (**Day 20**) doesn't? Putting two identical triangles together on the tray (**[BUILDER]**) always forms a Day 20 parallelogram! You can use **[f] [RECT] 2 [÷]** (**[SPARK]**) OR the 1-click **[g] [TRI]** key (**[STACK-HACK]**)!

Method	Keystroke Sequence	Snapshot & Result
Way 1: Dedicated **[g] [TRI]** Key	**10 [ENTER] 6 [g] [TRI]** (**[g][7]**)	**Y: 10 | X: 6** -> **30 cm²** in one press!
Way 2: Rectangle Cut in Half	**10 [ENTER] 6 [f] [RECT] 2 [÷]**	Rectangle is **60 cm²** -> cut in half = **30 cm²**!
Way 3: Half the Base First	**10 [ENTER] 2 [÷] 6 [×]**	Half of 10 is **5** -> 5 × 6 = **30 cm²**!

Problem 1 (Sailboat Triangle): A triangular sail has a base of 14 ft and a height of 18 ft. Find its area using [g] [TRI] and verify by cutting the rectangle in half.

1. [BLOCKS] TOUCH IT: **[ ]** Half of a 14 × 18 rectangle.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**14 [ENTER] 18 [g] [TRI]**): Snapshot **Y: 14 | X: 18** -> **126 ft²**.	3. [PROVE] VERIFIED SOLUTION: Doubling **126** gives the full rectangle **252**, and dividing by **18** restores base **14**!

Problem 2 (Right Triangle Skateboard Ramp): The side profile of a skate ramp has a flat base of 9 ft, a vertical back height of 4 ft, and a riding slope of 9.85 ft. What is the area of the side panel?

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**9 [ENTER] 4 [g] [TRI]**): Ignore the slanted riding slope **9.85**! Snapshot **Y: 9 | X: 4** -> **18 ft²**.	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

Problem 3 (Boss Level — Missing Triangle Height): A triangular pennant flag has an area of 60 in^2 and a base of 15 in. What is its height? (Remember: double the area to rebuild the full rectangle first!)

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**60 [ENTER] 2 [×] 15 [÷]**): Full rectangle area is **120** (**Y: 120 | X: 15**) -> **8 inches**. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[STACK-HACK]** Double-Area Then Divide): **60 [ENTER] 2 [×] 15 [÷]** = **8 in**. Way B (**[SPARK] [g] [TRI]** Key Verification): **15 [ENTER] 8 [g] [TRI]** = **60 in²**!	3. [PROVE] VERIFIED SOLUTION: Returns **60**!

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[STACK-HACK]** Double-Area Then Divide): **60 [ENTER] 2 [×] 15 [÷]** = **8 in**. Way B (**[SPARK] [g] [TRI]** Key Verification): **15 [ENTER] 8 [g] [TRI]** = **60 in²**!

Day 22 (Lesson 5.3): Trapezoids & Composite Figures Using the [⊠+] Running Ledger

SPIRAL PATH: Thread D (Inside-Out Chunking) • [[STACK-HACK] VISIT 3: STACK HACKER WAY — Boss Revisit of Days 3, 20 & 21] // VARIETY MODE: [TARGET] Mode B: Multi-Room Memory Vault ({{key:⊠+}} & {{key:⊠x}})

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Press **[g] [CL⊠]** (**[g][÷]**). Toss a rectangle **4 [ENTER] 5 [f] [RECT] [⊠+]** (**20**) and a triangle **4 [ENTER] 3 [g] [TRI] [⊠+]** (**6**) into memory. Press **[f] [⊠x]** (**[f][7]**) -> **26** total area! (**[ ] +5 Spark XP!**)

The Real Question: How do you find the area of a weird L-shaped room or a trapezoid without losing track of 4 different mini-calculations on scratch paper?

Why Should I Care? (2-Min Hook): Kids hate "Composite Figures" (weird shapes made of rectangles and triangles stuck together) and Trapezoids because you have to calculate 2 or 3 separate areas, write them all down, and add them at the end. On the SC-6, you have a Running Area Adder (**[⊠+]**)! Every time you finish one chunk of the shape, tap **[⊠+]** to toss it into the statistical bucket (**n = 1**, **n = 2**, **n = 3**...), then tap **[f] [⊠x]** (**[f][N]**) to read the grand total!
Why Try a New Way Today? (When the Old Way Gets Annoying): Memorizing the trapezoid formula is annoying—and what if an L-shape has 4 rooms? Instead of writing a giant equation, chop any weird shape into simple rectangles (**Day 20**) and triangles (**Day 21**) and toss each piece into **[⊠+]** (**[STACK-HACK]**)!

Problem 1 (Trapezoid Desk Top): A gaming desk is shaped like a trapezoid with parallel front/back edges b_1 = 48 in and b_2 = 36 in, and depth h = 24 in. Find its area.

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**48 [ENTER] 36 [+] 2 [÷] 24 [×]**): Snapshot **Y (Avg Base): 42 | X (Height): 24** -> **1008 in²**.	3. [PROVE] VERIFIED SOLUTION: Restores average base **42**!

Problem 2 (House Wall Composite — Rectangle + Roof Triangle): A shed wall is a 12 ft by 8 ft rectangle topped by a roof triangle with base 12 ft and height 5 ft. Use [⊠+] to find the total wall area.

1. [BLOCKS] TOUCH IT: **[ ]** Chunk 1 = Rectangle (12 × 8 = 96); Chunk 2 = Triangle (1/2 × 12 × 5 = 30).	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: **[g] [CL⊠]** -> **12 [ENTER] 8 [f] [RECT] [⊠+]** (**n = 1**) -> **12 [ENTER] 5 [g] [TRI] [⊠+]** (**n = 2**) -> **[f] [⊠x]** = **126 ft²**.	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

Problem 3 (Boss Level — Frame with Cutout Hole): A rectangular picture frame is 10 in by 8 in on the outside, with a 6 in by 4 in glass cutout in the middle. Use [⊠+] for the outside and [f] [⊠-] ([f][⊠+]) to subtract the hole!

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: **[g] [CL⊠]** -> **10 [ENTER] 8 [f] [RECT] [⊠+]** (+80) -> **6 [ENTER] 4 [f] [RECT] [f] [⊠-]** (-24) -> **[f] [⊠x]** = **56 in²**. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[STACK-HACK]** Outer Minus Cutout Hole): **10 [ENTER] 8 [f] [RECT]** (**80**) minus **6 [ENTER] 4 [f] [RECT]** (**24**) = **56 in²**. Way B (**[BUILDER]** 4 Border Strips Sum via **[⊠+]**): 20 + 20 + 8 + 8 = **56 in²**!	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[STACK-HACK]** Outer Minus Cutout Hole): **10 [ENTER] 8 [f] [RECT]** (**80**) minus **6 [ENTER] 4 [f] [RECT]** (**24**) = **56 in²**. Way B (**[BUILDER]** 4 Border Strips Sum via **[⊠+]**): 20 + 20 + 8 + 8 = **56 in²**!

Day 23 (Lesson 5.4): Coordinate Polygons & Planetary Clock Base-60 Time ([CLK])

SPIRAL PATH: Thread A & C Boomerang • [[BOOMERANG] VISIT 4: CROSS-CHAPTER BOOMERANG — Revisits Day 1 Modulo-60 & Day 10 Decimals] // VARIETY MODE: [SPARK] Mode C: Planetary Clock Gear Spin ({{key:CLK}})

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Spin the minute hand on your Planetary Clock (**TIME-01**) by 30 minutes (1/2 turn). Type **1.30 [f] [⊠HR]** (**[g][9]**) on the SC-6—watch **1 hr 30 min** turn into **1.5 hours**! (**[ ] +5 Spark XP!**)

The Real Question: How do you measure rectangles on a coordinate grid by subtracting coordinates—and how does the SC-6 Clock key (**[CLK]**) add and subtract hours and minutes without base-10 mistakes?

Why Should I Care? (2-Min Hook): If a movie starts at **1:45** and lasts **1 hour 35 minutes**, you can't type **1.45 + 1.35 = 2.80** on a normal calculator because there are 60 minutes in an hour, not 100! The Classroom Planetary Clock (**TIME-01**) uses a real 12:1 gear train, and the SC-6 **[CLK]** key calculates directly in **H.MMSS** base-60 time!
Why Try a New Way Today? (When the Old Way Gets Annoying): Subtracting time on paper (**1:20 PM - 11:45 AM**) fails if you borrow **100** instead of **60**. Spinning the Planetary Geared Clock (**[BUILDER]**) shows the 60-minute gear ratio physically, while **[CLK]** and **[f] [⊠HR]** (**[STACK-HACK]**) convert **3 hr 15 min** into **3.25 hr** automatically!

Camera Angle	What You Do	What You See
Angle 1: Physical Planetary Clock (**TIME-01**)	Set the hands to **1:45**. Rotate the minute hand forward **1** full turn (+1 hr -> 2:45) plus **35** minute ticks.	As the minute hand crosses **12**, the planetary gears carry **60** minutes into the next hour -> **3:20**!
Angle 2: SC-6 Modulo-60 Clock Mode (**[CLK]**)	Press **[CLK]** (**H.MM** mode). Key **1.45 [ENTER] 1.35 [+]**.	Screen displays **3.20** (3 hours, 20 minutes)—automatically bundling 60 minutes into 1 hour!
Angle 3: Decimal Hours Converter (**[f] [⊠HR]** / **[g] [⊠HMS]**)	Press **[f] [⊠HR]** (**[f][CLK]**) on **1.45** to turn **1 hr 45 min** into **1.75** hours (13/4 hr), or **[g] [⊠HMS]** (**[g][CLK]**) to convert back!	Connects fractions (45/60 = 3/4), decimals (0.75), and clock time (**:45**) in one click!

Problem 1 (Coordinate Box Dimensions): A rectangle on a grid has corners at (2, 3), (9, 3), (9, 8), and (2, 8). Find its width, height, perimeter, and area.

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: Width: **9 [ENTER] 2 [-]** (-> 7). Height: **8 [ENTER] 3 [-]** (-> 5). Snapshot **Y: 7 | X: 5**. Press **[f] [RECT]** -> Area = **35 sq units** (Perimeter = 7+7+5+5 = 24).	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

Problem 2 (3D Print Job Elapsed Time): A 3D print starts at `9:40` AM and takes `2 hours 50 minutes`. What time does it finish?		
1. [BLOCKS] TOUCH IT: `[ ]` Spun Planetary Clock from **`9:40`** forward **`2h 50m`**.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**`[CLK] 9.40 [ENTER] 2.50 [+]`**): Snapshot **`Y: 9.40 | X: 2.50`** -> **`12.30`** (**`12:30 PM`**). Return to normal numbers after via **`[NUM]`**!	3. [PROVE] VERIFIED SOLUTION: Subtracts **`2.50`** and returns to **`9.40`**!

Problem 3 (Boss Level — Hours to Decimal Paycheck): You tutor math for 3 hours and 15 minutes (`3.15` in `H.MM`) at $16.00 per hour. Convert `3.15` to decimal hours (`[f] [⌧HR]`) and multiply by `16`.		
1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**`3.15 [f] [⌧HR] 16 [×]`**): **`3.15`** converts to **`3.25 hours`** (**`Y: 3.25 | X: 16`**) -> **`52.00`**. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**`[STACK-HACK] [f] [⌧HR]`** Paycheck Shortcut): **`3.15 [f] [⌧HR]`** = **`3.25 hr`** × 16 = 52.00. Way B (**`[BUILDER]`** Split Hours + Quarter-Hour from Day 6): 3($16) = 48 plus 1/4($16) = 4 -> 52.00!	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.
[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**`[STACK-HACK] [f] [⌧HR]`** Paycheck Shortcut): **`3.15 [f] [⌧HR]`** = **`3.25 hr`** × 16 = 52.00. Way B (**`[BUILDER]`** Split Hours + Quarter-Hour from Day 6): 3(16) = 48 plus 1/4($16) = 4 -> 52.00!		

CHAPTER 6: Integers, Rational Numbers & The Coordinate Plane (Days 24–28)

Aligned to Big Ideas Math Grade 6, Chapter 6: Positive & Negative Integers, Comparing & Ordering Rational Numbers, Absolute Value, and the 4-Quadrant Coordinate Plane.

Day 24 (Lesson 6.1): Positive & Negative Integers — Bicolor Counters & The [+/-] Sign Flipper

SPIRAL PATH: Thread E (Zero-Pair & Distance) • [[BUILDER] VISIT 1: BUILDER WAY — Launch of Thread E] // VARIETY MODE: DICE Mode E: Zero-Pair Battle Royale ({{key:+/-}})

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Put 3 Yellow (+1) and 3 Red (-1) counters on your tray—each Yellow+Red pair cancels to 0! Now type **3 [ENTER] 3 [+/-] [+]** -> **0**! (**[ ] +5 Spark XP!**)

The Real Question: What is the difference between the subtraction action key **[-]** and the negative sign-flipper key **[+/-]**?

Why Should I Care? (2-Min Hook): On regular calculators, kids constantly get **SYNTAX ERROR** because they mix up the minus button and the negative button. On the SC-6, the difference is physical:
Why Try a New Way Today? (When the Old Way Gets Annoying): Negative numbers feel abstract until you hold Yellow (+1) and Red (-1) counters (**[BUILDER]**) that physically cancel in Zero Pairs, paired with the SC-6's dedicated **[+/-]** sign-flipper key (**[STACK-HACK]**)!

Real-World Action	Bicolor Counter Build (BASE-10)	SC-6 Keystrokes	Opposite ([+/-] Again)
Dive 15 feet below sea level	15 Red (-) counters	**15 [+/-]** -> **-15**	Tap **[+/-]** -> **+15**
Gain 25 XP points	25 Yellow (+) counters	**25** -> **25**	Tap **[+/-]** -> **-25**
Sea Level / Par / Break-Even	Equal Red & Yellow counters (Zero Pairs!)	**15 [+/-] [ENTER] 15 [+]** -> **0**	**0** is its own opposite!

Problem 1 (Opposite Flipper): Enter -42 on the SC-6, find its opposite by pressing [+/-], and find the opposite of the opposite!

1. [BLOCKS] TOUCH IT: **[ ]** Flipped counter from Red (-) to Yellow (+) and back to Red (-).	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**42 [+/-]**): Shows **-42**. Tap **[+/-]** -> **42** (Opposite). Tap **[+/-]** -> **-42** (-(-42) = -42... wait: the opposite of -42 is +42, and -(-42) = +42!).	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

Problem 2 (Zero-Pair Cancellation): A submarine is at -18 meters and rises +18 meters. Where is it now?

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**18 [+/-] [ENTER] 18 [+]**): Snapshot **Y: -18 | X: 18** -> **0** (Sea Level).	3. [PROVE] VERIFIED SOLUTION: Returns to **-18**!

Problem 3 (Boss Level — Bank Account Ledger): You start at $0, deposit $35, spend $50 (50 [+/-]), and deposit $15. What is your final balance?

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**35 [ENTER] 50 [+/-] [+] 15 [+]**): After first **[+]**: **-15**. Snapshot before final **[+]**: **Y: -15 | X: 15** -> **0.00**. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[STACK-HACK]** Running Stack Ledger): **35 [ENTER] 50 [+/-] [+] 20 [+]** = **+5**. Way B (**[SPARK]** Group Positives vs Negatives): Deposits ($35+20=55) minus Spend ($50) = **+$5**!	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[STACK-HACK]** Running Stack Ledger): **35 [ENTER] 50 [+/-] [+] 20 [+]** = **+5**. Way B (**[SPARK]** Group Positives vs Negatives): Deposits (35+20=55) minus Spend ($50) = **+$5**!

Day 25 (Lesson 6.2): Comparing & Ordering Integers and Rational Numbers on the Number Line

SPIRAL PATH: Thread E (Zero-Pair & Distance) • [[SPARK] VISIT 2: PATTERN WAY — Revisits Day 14 Ordering & Day 24 Integers] // VARIETY MODE: [DETECTIVE] Mode A: Spot the Glitch ('Is -18 Bigger Than -5?')

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Which is warmer: -5° or -18°? Type **5 [+/-] [ENTER] 18 [+/-] [f] [>]** (**[f][+]**)—the SC-6 outputs **1** (TRUE)! -5 is to the right of -18, so -5 > -18! (**[ ] +5 Spark XP!**)

The Real Question: Why is -3 GREATER than -8 even though 8 is bigger than 3—and how do the SC-6 comparison keys (**[f] [>]**, **[g] [<]**) test inequalities for you?

Why Should I Care? (2-Min Hook): When kids see -8 and -3, their instinct says -8 is bigger. Think about temperature or owing money: would you rather it be -3°F outside or -8°F? Would you rather owe your friend $3 (**-3**) or owe $8 (**-8**)? On a number line (**[f] [LINE]**, **[f][10F]**), further to the RIGHT is ALWAYS greater!
Why Try a New Way Today? (When the Old Way Gets Annoying): On the negative side of zero, bigger digits mean colder/smaller numbers (-18 < -5), which tricks your brain! Sliding beads on the Number Rack (**[BUILDER]**), checking **[f] [LINE]** (**[SPARK]**), and testing with **[g] [<]**/**[f] [>]** (**[STACK-HACK]**) gives you 3 ways to never get tricked!

Problem 1 (Freezer Showdown): Which is warmer (greater): -12°C or -5°C?

1. [BLOCKS] TOUCH IT: **[ ]** Located -5 to the right of -12 on the Number Line (**[f] [LINE]**).	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**12 [+/-] [ENTER] 5 [+/-] [g] [<]**): Snapshot **Y: -12 | X: -5** -> **-12 < -5** (or **-5 > -12**).	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

Problem 2 (Negative Fraction vs. Decimal): Compare -4/5 and -0.75. Which one is greater?

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**[.] 4 [.] 5 [+/-] [ENTER] [FDISP]**): Converts -4/5 to **-0.80**. Compare **Y: -0.80 | X: -0.75**. Since -0.75 is closer to zero, **-0.75 > -4/5**.	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

Problem 3 (Boss Level — Line Up From Least to Greatest): Order -1.5, 11/4, -13/4, and -0.5 from least (coldest/leftmost) to greatest (warmest/rightmost).

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (Convert to Decimals): **-1.50**, **+1.25**, **-1.75**, **-0.50**. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[BUILDER]** Number Line Position **[f] [LINE]**): Left-to-right order is **-1 3/4, -1.5, -0.5, 1 1/4**. Way B (**[STACK-HACK]** Pairwise **[g] [<]** Truth Check): **1.75 [+/-] [ENTER] 1.5 [+/-] [g] [<]** -> **1** (True)!	3. [PROVE] VERIFIED SOLUTION: **-1 3/4 (-1.75) < -1.5 < -0.5 < 1 1/4 (+1.25)**.

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[BUILDER]** Number Line Position **[f] [LINE]**): Left-to-right order is **-1 3/4, -1.5, -0.5, 1 1/4**. Way B (**[STACK-HACK]** Pairwise **[g] [<]** Truth Check): **1.75 [+/-] [ENTER] 1.5 [+/-] [g] [<]** -> **1** (True)!

Day 26 (Lesson 6.3): Absolute Value — Distance From Zero Is Always Positive (**[|x|]**)

SPIRAL PATH: Thread E (Zero-Pair & Distance) • [[STACK-HACK] VISIT 3: STACK HACKER WAY — Boss Revisit of Days 24 & 25] // VARIETY MODE: [SPARK] Mode C: Positive Mirror Test ([|x|] vs {{key:+/-}})

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Walk 6 steps backward (-6). How many steps did your feet take? 6 steps (you can't walk negative steps!). Type **6 [+/-]** and press **[|x|]** (**[f][+/-]**) -> **6**! (**[ ] +5 Spark XP!**)

The Real Question: Why does |-9| equal +9, while the opposite of +9 is -9?

Why Should I Care? (2-Min Hook): Students constantly confuse Opposite (**[+/-]**) with Absolute Value (**[|x|]**, **[g][+/-]**):
Why Try a New Way Today? (When the Old Way Gets Annoying): Counting steps to zero on a number line (**Day 25**) is easy for |-4|, but in multi-step expressions like 50 - |-18|, kids confuse 'opposite' (**[+/-]**) with 'distance from zero' (**[|x|]**). Using **[|x|]** on Floor 1 (**X**) while **50** waits on Floor 2 (**Y**) keeps them crystal clear!

Expression	What It Means in Plain English	SC-6 Keystrokes	Result
|-14|	How far is -14 from 0?	**14 [+/-] [g] [|x|]** (**[g][+/-]**)	**14**
|14|	How far is +14 from 0?	**14 [g] [|x|]** (**[g][+/-]**)	**14**
-|-14|	Find **|-14|** first (**14**), THEN stick a negative sign outside!	**14 [+/-] [g] [|x|] [+/-]**	**-14**

Problem 1 (Step Distance Check): Evaluate |-27| + |13|.		
1. [BLOCKS] TOUCH IT: `[ ]` 27 steps from zero plus 13 steps from zero.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**`27 [+/-] [g] [|x|] [ENTER] 13 [g] [|x|]`**): Snapshot before **`[+]`**: **`Y: 27 | X: 13`**. Press **`[+]`** -> **`40`**.	3. [PROVE] VERIFIED SOLUTION: Restores **`27`**!

Problem 2 (Submarine vs. Seagull): A scuba diver is at -32 ft and a seagull is at +25 ft. Which one is closer to sea level (0), and by how many feet?		
1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**`32 [+/-] [g] [|x|] [ENTER] 25 [g] [|x|]`**): Snapshot **`Y (Diver dist): 32 | X (Gull dist): 25`**. Press **`[-]`** -> **`7 ft`**. The seagull is closer to sea level (25 ft vs. 32 ft)!	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

Problem 3 (Boss Level — Outside Negative Trap): Evaluate 50 - |-18|. (Careful: does |-18| turn into +18 BEFORE you subtract from 50?)		
1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**`50 [ENTER] 18 [+/-] [g] [|x|]`**): Snapshot before **`[-]`**: **`Y: 50 | X: 18`**. Press **`[-]`** -> **`32`** (NOT 68!). • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**`[STACK-HACK]`** Elevator Stack): **`50 [ENTER] 18 [+/-] [|x|]`** (**`Y: 50 | X: 18`**) **`[-]`** = 32. Way B (**`[BUILDER]`** Distance-First Rewrite): Distance from -18 to 0 is +18, so 50 - 18 = 32!	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.
[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**`[STACK-HACK]`** Elevator Stack): **`50 [ENTER] 18 [+/-] [|x|]`** (**`Y: 50 | X: 18`**) **`[-]`** = 32. Way B (**`[BUILDER]`** Distance-First Rewrite): Distance from -18 to 0 is +18, so 50 - 18 = 32!		

Day 27 (Lesson 6.4): The Four-Quadrant Coordinate Plane & Axis Reflections

`SPIRAL PATH: Thread E (Zero-Pair & Distance) • [[SPARK] VISIT 2: PATTERN WAY — Revisits Day 24 Sign Flipping in 2D] // VARIETY MODE: [TARGET] Mode B: 4-Quadrant Mirror Treasure Hunt`

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Start at (3, 4) in Quadrant I. Flip the x-sign with **`[+/-]`** to (-3, 4)—you just jumped across the y-axis into Quadrant II! (**`[ ] +5 Spark XP!`**)

The Real Question: How do you remember whether (x, y) goes sideways or up-and-down first—and what happens to the signs when a point reflects across an axis?

Why Should I Care? (2-Min Hook): Every 2D video game map, Minecraft coordinate (**`X, Z`**), and 3D printer bed uses a (x, y) coordinate grid. To never mix up x and y, remember the Elevator Rule: You have to walk ACROSS the lobby (x, left/right) BEFORE you can ride UP/DOWN the elevator (y)!
Why Try a New Way Today? (When the Old Way Gets Annoying): Plotting 4 quadrants on paper gets messy when you mix up (x,y) order or reflection signs. Snapping pegs on the Coordinate Baseplate (**`[BUILDER]`**) shows that reflecting across an axis is just pressing **`[+/-]`** (**`[STACK-HACK]`**) on one coordinate!

Quadrant / Reflection	Signs of (x, y)	Example from (4, 3)	SC-6 Sign-Flip Rule (**`[+/-]`**)
Quadrant I (Top-Right)	(+, +)	(4, 3)	Starting point: **`Y: 4 | X: 3`**
Reflect Across y-Axis -> Quad II (Top-Left)	(-, +)	(-4, 3)	Only x flips (**`[+/-]`**)! Height y=3 stays the same!
Reflect Across x-Axis -> Quad IV (Bottom-Right)	(+, -)	(4, -3)	Only y flips (**`[+/-]`**)! Horizontal x=4 stays the same!
Reflect Across Both Axes -> Quad III (Bottom-Left)	(-, -)	(-4, -3)	Both x and y flip (**`[+/-]`**)!

Problem 1 (Quadrant Detective): Without drawing a grid, name the Quadrant (I, II, III, or IV) for A(-6, 5), B(4, -7), and C(-3, -8).		
1. [BLOCKS] TOUCH IT: `[ ]` Placed tiles on the 4-quadrant desk cross.	2. [STACK] KEYSTROKES & Y|X: Verified on 4-line XYZT stack.	3. [PROVE] VERIFIED SOLUTION: A(-,+) -> Quadrant II; B(+,-) -> Quadrant IV; C(-,-) -> Quadrant III.

Problem 2 (Mirror Reflection): Start at point P(-5, 8). Reflect it across the x-axis first, then reflect that new point across the y-axis. Where do you land?

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: Across x-axis flips the y-coordinate (**`8 [+/-]`** -> -8): **`(-5, -8)`**. Across y-axis flips the x-coordinate (**`5 [+/-] [+/-]`** -> +5): **`(5, -8)`**.	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

Problem 3 (Boss Level — On the Fence): Which axis does (0, -9) sit on, and which axis does (-7, 0) sit on?

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**`[BUILDER]`** Baseplate Peg Walk): (0,-9) goes 0 sideways, 9 down (y-axis); (-7,0) goes 7 left, 0 up/down (x-axis). Way B (**`[SPARK]`** Zero-Coordinate Rule): x=0 -> y-axis; y=0 -> x-axis!	3. [PROVE] VERIFIED SOLUTION: (0, -9) goes 0 sideways and -9 down -> **`y-axis`**. (-7, 0) goes -7 left and 0 up/down -> **`x-axis`**.

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**`[BUILDER]`** Baseplate Peg Walk): (0,-9) goes 0 sideways, 9 down (y-axis); (-7,0) goes 7 left, 0 up/down (x-axis). Way B (**`[SPARK]`** Zero-Coordinate Rule): x=0 -> y-axis; y=0 -> x-axis!

Day 28 (Lesson 6.5): Distance Between Points on the Coordinate Plane

`SPIRAL PATH: Thread E & D Boomerang • [[BOOMERANG] VISIT 4: CROSS-CHAPTER BOOMERANG — Revisits Day 20 Perimeter & Day 26 |x|] // VARIETY MODE: [BOOMERANG] Mode D: Bridge Builder Distance Race ([|x|])`

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): How far is -3 from +4? It's 3 steps to zero plus 4 steps past zero = **7**! Now type **`3 [+/-] [ENTER] 4 [-] [|x|]`** -> **7** in 2 clicks! (**`[ ] +5 Spark XP!`**)

The Real Question: When finding the distance between (-6, 4) and (9, 4), why do you ADD |-6| + |9| = 15 when they are in different quadrants, but SUBTRACT when they are in the same quadrant?

Why Should I Care? (2-Min Hook): Imagine walking from 6 blocks West of Main Street (**-6**) to 9 blocks East of Main Street (**+9**). First you walk 6 blocks to reach Main Street (**0**), and then you walk 9 more blocks East (6 + 9 = 15 blocks)!
Why Try a New Way Today? (When the Old Way Gets Annoying): Counting grid squares one by one works when points are 4 units apart, but what if a bridge spans from (-45, 12) to (80, 12)? Using Day 26's absolute value distance (**`[|x|]`**) finds the span across any two coordinates in 2 clicks!

Point Pair	Keystroke Sequence	Before `[|x|]`	After `[g] [|x|]` (Distance!)				
(-6, 4) to (9, 4) (Crosses Axis)	**`6 [+/-] [ENTER] 9 [-] [g] [|x|]`**	**-15**	**15** units ($	-6	+	9	= 15$)
(-12, -3) to (-4, -3) (Same Side)	**`12 [+/-] [ENTER] 4 [+/-] [-] [g] [|x|]`**	**-8**	**8** units ($	-12	-	-4	= 8$)

Problem 1 (Cross-Axis Bridge): Find the distance between (-8, 5) and (11, 5).

1. [BLOCKS] TOUCH IT: **`[ ]`** 8 steps to the axis + 11 steps past the axis.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**`8 [+/-] [g] [|x|] [ENTER] 11 [g] [|x|] [+]`**): Snapshot **`Y: 8 | X: 11`** -> **`19 units`**.	3. [PROVE] VERIFIED SOLUTION: Also gives **19**!

Problem 2 (Same-Quadrant Hop): Find the vertical distance between (-3, -4) and (-3, -15).

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**`15 [+/-] [g] [|x|] [ENTER] 4 [+/-] [g] [|x|] [-]`**): Snapshot **`Y: 15 | X: 4`** -> **`11 units`**.	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

Problem 3 (Boss Level — 4-Quadrant Treasure Map Perimeter): A rectangle has vertices at (-4, 6), (5, 6), (5, -2), and (-4, -2). Find its width, height, perimeter, and area!		
1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: Width: |-4| + |5| = 9. Height: |6| + |-2| = 8. Snapshot **`Y: 9 | X: 8`**. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**`[BUILDER]`** Add Distances to Axis): Width = |-4| + |5| = 9; Height = |6| + |-2| = 8 -> P = 2(9+8) = 34. Way B (**`[STACK-HACK]`** Coordinate Subtraction + **`[|x|]`**): **`|-4 - 5| = 9`** and **`|6 - (-2)| = 8`** -> Perimeter = 34!	3. [PROVE] VERIFIED SOLUTION: Area (**`9 [ENTER] 8 [f] [RECT]`**) = **`72 sq units`**; Perimeter (9+9+8+8) = **`34 units`**.
[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**`[BUILDER]`** Add Distances to Axis): Width = |-4| + |5| = 9; Height = |6| + |-2| = 8 -> P = 2(9+8) = 34. Way B (**`[STACK-HACK]`** Coordinate Subtraction + **`[|x|]`**): **`|-4 - 5| = 9`** and **`|6 - (-2)| = 8`** -> Perimeter = 34!		

CHAPTER 7: Equations & Inequalities (Days 29–33)

Aligned to Big Ideas Math Grade 6, Chapter 7: Writing Equations, Solving One-Step Equations, Writing & Graphing Inequalities, and Independent/Dependent Variables.

Day 29 (Lesson 7.1): Writing Equations & Testing Solutions — The Balance Scale Rule

SPIRAL PATH: Thread F (Balance & Fair Share) • [[BUILDER] VISIT 1: BUILDER WAY — Launch of Thread F] // VARIETY MODE: [DETECTIVE] Mode A: Balance Scale Lie Detector ({{key:≠}})

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Does x = 5 solve x + 4 = 9? Type **5 [ENTER] 4 [+]** (**9**), then **9 [≠]** (**[g][-]**). The screen says **0** (Zero difference = Perfectly Balanced)! (**[ ] +5 Spark XP!**)

The Real Question: What does the **=** sign actually mean—and how can you tell in 3 seconds if a number is the true solution to an equation?

Why Should I Care? (2-Min Hook): From Grade 1, kids think **=** means "Write the answer next." In algebra, **=** does NOT mean "here comes the answer"—it is the center pivot of a Balance Scale! Whatever sits on the Left Tray must weigh the exact same as whatever sits on the Right Tray. And on the SC-6, Floor 2 (**Y**) is the Left Tray and Floor 1 (**X**) is the Right Tray!
Why Try a New Way Today? (When the Old Way Gets Annoying): Before learning formal equation-solving steps, you can test whether a number is the solution just by dropping it onto the Algebra Stage (**[BUILDER]**) or checking **[≠]** (**[STACK-HACK]**) to see if both sides weigh the exact same!

Camera Angle	What You Do	What You See
Angle 1: Physical Algebra Stage (**ALGB-01**)	Place three x-rods on the left side of the stage and **19** unit cubes on the right.	If each x-rod equals **8** (3 × 8 = 24), taking away **5** leaves **19**—balanced!
Angle 2: SC-6 **Y** vs. **X** Balance Test	Put the Right Side (**19**) on Floor 2 (**Y**: **19 [ENTER]**). Now evaluate the Left Side with x = 8 (**3 [ENTER] 8 [×] 5 [-]**).	Look at your screen: **Y: 19 | X: 19**! When Floor 2 and Floor 1 match, the scale is balanced!
Angle 3: Word-to-Math Translator	"5 less than 3 times a number is 19" -> 3x - 5 = 19.	"is" / "equals" / "totals" = the balance point **=**!

Problem 1 (Balance Scale Solution Tester): Test whether x = 7 or x = 9 solves 4x + 5 = 41.

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: Park **41 [ENTER]**. Test x = 9: **4 [ENTER] 9 [×] 5 [+]** -> Snapshot **Y: 41 | X: 41**!	3. [PROVE] VERIFIED SOLUTION: x = 9 balances both floors at **41**!

Problem 2 (Translate Words to Tiles): A squad buys 4 identical pizza pies plus a $6 delivery fee for a total of $54. Write the equation and find the price p of one pie.

1. [BLOCKS] TOUCH IT: **[ ]** Four p-rods + 6 unit cubes = 54 cubes (-> 4p + 6 = 54).	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**54 [ENTER] 6 [-] 4 [÷]**): Snapshot before **[÷]**: **Y: 48 | X: 4** -> **p = $12**.	3. [PROVE] VERIFIED SOLUTION: Returns **54**!

Problem 3 (Boss Level — "Less Than" Flip Trap): Translate "8 less than a number n is 25" into an equation and solve it. (Careful: "8 less than n" means start with n and take 8 away: n - 8 = 25, NOT 8 - n!)

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**25 [ENTER] 8 [+]**): Snapshot **Y: 25 | X: 8** -> **n = 33**. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[STACK-HACK]** Inverse Undo): Equation is n - 8 = 25 -> **25 [ENTER] 8 [+]** = 33. Way B (**[BUILDER]** Plug-In Balance Proof from Day 16): Test n=33 -> 33 - 8 = 25 (Balanced!)!	3. [PROVE] VERIFIED SOLUTION: 33 - 8 = 25!

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[STACK-HACK]** Inverse Undo): Equation is n - 8 = 25 -> **25 [ENTER] 8 [+]** = 33. Way B (**[BUILDER]** Plug-In Balance Proof from Day 16): Test n=33 -> 33 - 8 = 25 (Balanced!)!

Day 30 (Lesson 7.2): Solving One-Step Addition & Subtraction Equations — Inverse Undo & [f] [LASTX]

SPIRAL PATH: Thread F & B Boomerang • [[SPARK] VISIT 2: PATTERN WAY — Revisits Day 8 Inverse & Day 29 Balance] // VARIETY MODE: [TARGET] Mode B: Mystery Box Unwrap + {{key:LASTX}} Bounce-Back

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): A mystery box plus 5 cubes weighs 12 cubes (x + 5 = 12). Take 5 cubes off both sides: **12 [ENTER] 5 [-]** -> **x = 7**! Now press **[f] [LASTX] [+]**—watch **12** bounce right back! (**[ ] +5 Spark XP!**)

The Real Question: Why do teachers make you write - 14 under both sides of x + 14 = 39 when you can already see the answer is 39 - 14?

Why Should I Care? (2-Min Hook): Kids with fast number sense hate one-step equations because teachers dock points if they don't write:
Why Try a New Way Today? (When the Old Way Gets Annoying): Guessing-and-checking (**Day 29**) works for easy whole numbers, but good luck guessing m + 3/8 = 5/6! Sweeping blocks off both pans (**[BUILDER]**) shows why the inverse operation (**[-]** or **[+]**) isolates the variable in 1 step, and **[f] [LASTX]** (**[STACK-HACK]**) proves it!

Equation Type	Physical Algebra Stage Action (ALGB-01)	SC-6 Inverse Stack Keystrokes	[f] [LASTX] 2-Click Self-Proof
x + 14 = 39	Remove **14** unit cubes from both pans so the x-rod stands alone!	**39 [ENTER] 14 [-]** -> **x = 25** (Snapshot **Y: 39 | X: 14**)	Press **[f] [LASTX] [+]** -> **39**!
x - 18 = 45	Put **18** unit cubes back onto both pans to cancel the missing 18!	**45 [ENTER] 18 [+]** -> **x = 63** (Snapshot **Y: 45 | X: 18**)	Press **[f] [LASTX] [-]** -> **45**!

Problem 1 (Undo Addition): Solve x + 28 = 73 and prove with [f] [LASTX].

1. [BLOCKS] TOUCH IT: **[ ]** Removed 28 cubes from both sides.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**73 [ENTER] 28 [-]**): Snapshot **Y: 73 | X: 28** -> **x = 45**.	3. [PROVE] VERIFIED SOLUTION: 45 + 28 = 73!

Problem 2 (Undo Subtraction): Solve w - 19.6 = 34.8 and prove with [f] [LASTX].

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**34.8 [ENTER] 19.6 [+]**): Snapshot **Y: 34.80 | X: 19.60** -> **w = 54.4**.	3. [PROVE] VERIFIED SOLUTION: 54.4 - 19.6 = 34.8!

Problem 3 (Boss Level — Fraction One-Step Equation): Solve m + 3/8 = 5/6 on the SC-6 and prove with [f] [LASTX].

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**[.] 5 [.] 6 [ENTER] [.] 3 [.] 8 [-]**): Snapshot **Y: 5/6 | X: 3/8** -> **m = 11/24**. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[STACK-HACK]** Fraction Stack Undo): **[.] 5 [.] 6 [ENTER] [.] 3 [.] 8 [-]** = **11/24**. Way B (**[SPARK]** Common-Denominator 24ths from Day 6): 20/24 - 9/24 = **11/24**!	3. [PROVE] VERIFIED SOLUTION: 11/24 + 3/8 = 5/6!

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[STACK-HACK]** Fraction Stack Undo): **[.] 5 [.] 6 [ENTER] [.] 3 [.] 8 [-]** = **11/24**. Way B (**[SPARK]** Common-Denominator 24ths from Day 6): 20/24 - 9/24 = **11/24**!

Day 31 (Lesson 7.3): Solving One-Step Multiplication & Division Equations — Equal Groups on the Stage

SPIRAL PATH: Thread F, A & B Boomerang • [[STACK-HACK] VISIT 3: STACK HACKER WAY — Boss Revisit of Days 8, 29 & 30] // VARIETY MODE: [BOOMERANG] Mode D: Twin-Method Showdown ({{key:÷}} vs {{key:1/x}})

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): 3 identical x-rods weigh 15 cubes (3x = 15). Split 15 into 3 equal piles: **15 [ENTER] 3 [÷]** -> **x = 5**! Press **[f] [LASTX] [×]** -> **15**! (**[ ] +5 Spark XP!**)

The Real Question: How do you undo multiplication (6x = 84), division (x/5 = 13), or a fraction coefficient (3/4x = 18) in one clean stack move?

Why Should I Care? (2-Min Hook): On the Algebra Stage (**ALGB-01**), 6x = 84 literally means 6 identical x-rods weigh 84 units total. To find the weight of one x-rod, you deal the 84 units into 6 equal piles (**84 ÷ 6 = 14**)! And if a fraction is stuck to x like 3/4x = 18, you either divide by 3/4 OR multiply by its reciprocal **[f] [1/x]** (4/3)!
Why Try a New Way Today? (When the Old Way Gets Annoying): When an equation has a fraction coefficient like 2/5n = 16, kids freeze up. Today you discover two awesome ways that give the exact same answer: dividing by 2/5 (**Way A**) OR using Day 8's Reciprocal Flip **[f] [1/x]** (**Way B**)!

Equation Type	What It Means Physically	SC-6 Inverse Stack Keystrokes	[f] [LASTX] 2-Click Self-Proof
6x = 84	6 equal x-rods total 84. Split into 6 groups!	**84 [ENTER] 6 [÷]** -> **x = 14**	**[f] [LASTX] [×]** -> **84**!
x/5 = 13	One-fifth of x is 13. Put 5 fifths together!	**13 [ENTER] 5 [×]** -> **x = 65**	**[f] [LASTX] [÷]** -> **13**!
3/4x = 18	3/4 of x is 18. Divide by 3/4 (or multiply by **1/x** = 4/3)!	**18 [ENTER] [.] 3 [.] 4 [÷]** -> **x = 24**	**[f] [LASTX] [×]** -> **18**!

Problem 1 (Undo Multiplication): Solve 8x = 136 and prove with [f] [LASTX].

1. [BLOCKS] TOUCH IT: **[ ]** Split 136 equally across 8 x-rods.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**136 [ENTER] 8 [÷]**): Snapshot **Y: 136 | X: 8** -> **x = 17**.	3. [PROVE] VERIFIED SOLUTION: 17 × 8 = 136!

Problem 2 (Undo Division): Solve k/7 = 14.5 and prove with [f] [LASTX].

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**14.5 [ENTER] 7 [×]**): Snapshot **Y: 14.50 | X: 7** -> **k = 101.5**.	3. [PROVE] VERIFIED SOLUTION: 101.5 ÷ 7 = 14.5!

Problem 3 (Boss Level — Fraction Coefficient Both Ways): Solve 2/5n = 16 using BOTH division ([÷]) and reciprocal multiplication ([f] [1/x] [×]).

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**16 [ENTER] [.] 2 [.] 5 [f] [1/x]**): Snapshot before **[×]**: **Y: 16 | X: 5/2 (2 1/2)**. Press **[×]** -> **n = 40**. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[STACK-HACK]** Direct Fraction Divide): **16 [ENTER] [.] 2 [.] 5 [÷]** = 40. Way B (**[SPARK]** Reciprocal Flip **[f] [1/x]** from Day 8): **16 [ENTER] [.] 2 [.] 5 [f] [1/x] [×]** (16 × 5/2) = 40!	3. [PROVE] VERIFIED SOLUTION: Returns **16**!

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[STACK-HACK]** Direct Fraction Divide): **16 [ENTER] [.] 2 [.] 5 [÷]** = 40. Way B (**[SPARK]** Reciprocal Flip **[f] [1/x]** from Day 8): **16 [ENTER] [.] 2 [.] 5 [f] [1/x] [×]** (16 × 5/2) = 40!

Day 32 (Lesson 7.4): Writing, Graphing & Solving Inequalities ([f] [>], [g] [<], [f] [≥], [g] [≤])

SPIRAL PATH: Thread F & E Boomerang • [[BOOMERANG] VISIT 4: CROSS-CHAPTER BOOMERANG — Revisits Day 25 Number Line & Day 29 Balance] // VARIETY MODE: [SPARK] Mode C: Coaster Height Truth Detector ({{key:≥}} & {{key:≤}})

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): You need at least 54 inches (h >= 54) to ride. Are 54 inches enough? Type **54 [ENTER] 54 [f] [≥]** ([f][×]) -> **1** (TRUE, Ride On!) ([ ] +5 Spark XP!)

The Real Question: How do you know whether the circle on the number line is OPEN (**o**) or SOLID (**✱**), which way the arrow points, and how to test your answer on the SC-6?

Why Should I Care? (2-Min Hook): Roller coaster height signs ("Must be at least 48 inches tall": h >= 48) and elevator weight limits ("At most 2,000 pounds": w <= 2000) are inequalities! Look at the bottom-right corner of your SC-6 (**[✱]** and **[+]**):
Why Try a New Way Today? (When the Old Way Gets Annoying): Inequalities (>, <, >=, <=) have infinitely many solutions! Combining Day 30–31's Balance Undo (**[SPARK]**) with Day 25's SC-6 Truth Detector keys (**[f] [≥]**, **[g] [≤] [STACK-HACK]**) lets you solve AND test boundary numbers in seconds!

Everyday Phrase	Symbol	Circle Type	SC-6 Test Key (= True, 0 = False)
"More than" / "Exceeds"	**>**	Open **o** -> Right	**[f] [>]** ([f][✱])
"Less than" / "Fewer than"	**<**	Open **o** -> Left	**[g] [<]** ([g][✱])
"At least" / "No less than" / "Minimum"	**≥**	Solid **✱** -> Right	**[f] [≥]** ([f][+])
"At most" / "No more than" / "Maximum"	**≤**	Solid **✱** -> Left	**[g] [≤]** ([g][+])

Problem 1 (Solve & Test Boundary + Direction): Solve x - 7 >= 12, tell whether the dot at 19 is Open or Solid, and test x = 20 on the SC-6!

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**12 [ENTER] 7 [+]**): Boundary is **x ≥ 19** (Solid dot **✱** at 19, arrow Right).	3. [PROVE] VERIFIED SOLUTION: **20 [ENTER] 7 [-]** (-> 13) **12 [f] [≥]** -> Screen displays **1** (TRUE!).

Problem 2 (Gaming Backpack Weight Limit): Your character can carry at most 60 pounds without slowing down. You already carry a 15-pound shield, and each gold bar weighs 3 pounds (3g + 15 <= 60). What is the maximum number of gold bars g you can carry?

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**60 [ENTER] 15 [-] 3 [÷]**): Snapshot **Y: 45 | X: 3** -> **g ≤ 15 bars** (Solid dot **✱** at 15, arrow Left).	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

Problem 3 (Boss Level — Inequality Truth Detector): Which numbers from the set {4, 5, 6} are solutions to 4n + 3 > 23?

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: Solve boundary: **23 [ENTER] 3 [-] 4 [÷]** -> n > 5 (Open circle **o** at 5!). • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[STACK-HACK]** Truth-Detector Plug-In): Test x=4,5,6 in 4x - 3 >= 17: at x=5, **17 >= 17** (**1 = True**). Way B (**[SPARK]** Balance Solve First): 4x >= 20 -> x >= 5, so **{5, 6}** work!	3. [PROVE] VERIFIED SOLUTION: At n = 5, 4(5)+3 = 23, and **23 [ENTER] 23 [f] [>]** returns **0** (FALSE!) because 23 is not strictly greater than 23! Only **6** is a solution!

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[STACK-HACK]** Truth-Detector Plug-In): Test x=4,5,6 in 4x - 3 >= 17: at x=5, **17 >= 17** (**1 = True**). Way B (**[SPARK]** Balance Solve First): 4x >= 20 -> x >= 5, so **{5, 6}** work!

Day 33 (Lesson 7.5): Two-Variable Relationships (y = kx and y = x + b) — Tables & Graphs

SPIRAL PATH: Thread C & F Boomerang • [[BOOMERANG] VISIT 5: CROSS-CHAPTER BOOMERANG — Revisits Day 12 REP, Day 16 Formulas & Day 27 Graphs] // VARIETY MODE: DICE Mode E: Function Machine Table Builder ({{key:REP}})

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Each level gives 6 gold coins (y = 6x). Type **6 [g] [REP]** and tap **[+]** three times: **6 -> 12 -> 18 -> 24**! You just generated 4 (x,y) graph points in 3 seconds! (**[ ] +5 Spark XP!**)

The Real Question: How do you tell right away whether a table of (x, y) numbers is an ADDITION rule (y = x + b) or a MULTIPLICATION rule (y = kx)—and how do you fill out the whole table in 5 seconds?

Why Should I Care? (2-Min Hook): Filling out input-output tables by hand is tedious. Here is the 2-second detector:
Why Try a New Way Today? (When the Old Way Gets Annoying): Today ties together three whole chapters: a two-variable equation like y = 12.5x is simultaneously a Day 12 Ratio Table (**[SPARK]**), a Day 16 Formula (**[STACK-HACK]**), and a Day 27 Coordinate Line (**[BUILDER]**)!

Representation	Additive Rule (y = x + 6)	Multiplicative Rule (y = 6x)
1. Physical Tile Build (**COUNT-01**)	Start with x tiles and drop **6** extra tiles on top every time.	Build x rows of **6** tiles each!
2. Table Pattern Detector	Row (1, 7), Row (2, 8), Row (3, 9): y - x is always **6**!	Row (1, 6), Row (2, 12), Row (3, 18): y ÷ x is always **6**!
3. Graph Shape	Straight line starting at (0, 6).	Straight line passing right through the origin (0, 0)!

Problem 1 (Rule Detective): A table has points (2, 14), (5, 35), and (8, 56). Is it y = x + b or y = kx? Find the equation and find y when x = 12.

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**14 [ENTER] 2 [÷] 35 [ENTER] 5 [÷]**): Snapshot **Y: 7 | X: 7** (Constant ratio k = 7!). Equation: **y = 7x**. For x = 12: **7 [ENTER] 12 [×]** = **84**.	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

Problem 2 (Coupon Discount Table): A table has points (10, 6), (15, 11), and (22, 18). Find the equation relating original price x and final price y.

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**10 [ENTER] 6 [-] 15 [ENTER] 11 [-]**): Snapshot **Y: 4 | X: 4** (Always $4 less!). Equation: **y = x - 4**.	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

Problem 3 (Boss Level — Independent vs. Dependent): You earn $12.50 per hour (h) mowing lawns, for total pay p = 12.50h. Which variable is Independent (the input you choose) and which is Dependent (the output that depends on it)? Find p for h = 6.5 hours.

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**12.50 [ENTER] 6.5 [×]**): Independent = **h** (hours); Dependent = **p** (pay). Total pay = **81.25**. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[STACK-HACK]** Inverse Equation Undo for p=100): **100 [ENTER] 12.5 [÷]** = 8 hours. Way B (**[SPARK]** Double-Up Ratio Table from Day 12): 1 hr=12.50 -> 2 hr=25 -> 4 hr=50 -> 8 hr= 100!	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[STACK-HACK]** Inverse Equation Undo for p=100): **100 [ENTER] 12.5 [÷]** = 8 hours. Way B (**[SPARK]** Double-Up Ratio Table from Day 12): 1 hr=12.50 -> 2 hr=25 -> 4 hr=50 -> 8 hr= $100!

CHAPTER 8: Surface Area & Volume of 3D Solids (Days 34–37)

Aligned to Big Ideas Math Grade 6, Chapter 8: Three-Dimensional Figures, Surface Area of Prisms & Pyramids, Volume of Rectangular Prisms, and Circle/Cone Geometry.

Day 34 (Lesson 8.1): 3D Solids, Nets & Surface Area of Rectangular Prisms Without Losing Count

SPIRAL PATH: Thread D (Inside-Out Chunking) • [[BOOMERANG] VISIT 4: CROSS-CHAPTER BOOMERANG — Revisits Day 20 RECT & Day 22 ⊠+] // VARIETY MODE: [TARGET] Mode B: Unfold the 6-Wall Box ({{key:RECT}} + {{key:⊠+}})

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Look at a 2 by 2 by 2 cube of blocks. Each square wall has area 2 × 2 = 4, and a cube has 6 walls (6 × 4 = 24)! Key **4 [ENTER] 6 [×]** -> **24**! (**[ ] +5 Spark XP!**)

The Real Question: How do you find the surface area of a 6-sided box without losing track of the 6 rectangles or forgetting to double the 3 pairs of faces?

Why Should I Care? (2-Min Hook): Surface area of a rectangular prism is a classic "show your work" nightmare: kids have to calculate 6 different rectangles (Top, Bottom, Front, Back, Left, Right), write them all down, and add them up—and 80% of mistakes happen from forgetting one face or adding wrong at the end!
Why Try a New Way Today? (When the Old Way Gets Annoying): Finding the surface area of a 3D box by writing 2(lw + lh + wh) on paper is a magnet for arithmetic mistakes. Unfolding the 6 faces (**[BUILDER]**) and feeding the 3 pairs into Day 22's **[f] [RECT]** + **[⊠+]** ledger (**[STACK-HACK]**) never drops a face!

Face Pair on Physical Box	Dimensions	SC-6 Keystrokes (Start with [g] [CL⊠] [g][<-])	Running Face Counter (X)
1. Top & Bottom Pair	Two 8 × 5 rectangles	**8 [ENTER] 5 [f] [RECT] 2 [×] [⊠+]** (80 cm²)	**n = 1**
2. Front & Back Pair	Two 8 × 3 rectangles	**8 [ENTER] 3 [f] [RECT] 2 [×] [⊠+]** (48 cm²)	**n = 2**
3. Left & Right Pair	Two 5 × 3 rectangles	**5 [ENTER] 3 [f] [RECT] 2 [×] [⊠+]** (30 cm²)	**n = 3** (All 3 pairs logged!)
4. Read Total Surface Area	Sum of all 6 faces	Press **[f] [⊠x]** (**[f][N]**)	**158 cm²**!

Problem 1 (3-Pair Box Ledger): Find the surface area of a rectangular prism measuring 6 cm by 4 cm by 3 cm.

1. [BLOCKS] TOUCH IT: **[ ]** Touched Top/Bottom (6× 4), Front/Back (6× 3), and Left/Right (4× 3) on box.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: **[g] [CL⊠]** -> Pair 1: **6 [ENTER] 4 [f] [RECT] 2 [×] [⊠+]** (**48**) -> Pair 2: **6 [ENTER] 3 [f] [RECT] 2 [×] [⊠+]** (**36**) -> Pair 3: **4 [ENTER] 3 [f] [RECT] 2 [×] [⊠+]** (**24**) -> **[f] [⊠x]** = **108 cm²**.	3. [PROVE] VERIFIED SOLUTION: Check **[N]** = **3** pairs (**48 + 36 + 24 = 108**)!

Problem 2 (Cube Shortcut): A cube has 6 identical square faces, each of side length 7 in (SA = 6s^2). Find its surface area!

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**7 [x²] 6 [×]**): Snapshot **Y (1 Face): 49 | X: 6** -> **294 in²**.	3. [PROVE] VERIFIED SOLUTION: Restores side length **7**!

Problem 3 (Boss Level — Open-Top Storage Bin): You are 3D printing an open-top Gridfinity bin (10 cm long, 8 cm wide, 5 cm tall). Because there is NO lid, there is only 1 bottom (10 × 8), plus 2 front/back walls and 2 left/right walls! Find the outer surface area.

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: **[g] [CL⊠]** -> Bottom (only 1!): **10 [ENTER] 8 [f] [RECT] [⊠+]** (**80**) -> Front/Back: **10 [ENTER] 5 [f] [RECT] 2 [×] [⊠+]** (**100**) -> Left/Right: **8 [ENTER] 5 [f] [RECT] 2 [×] [⊠+]** (**80**) -> **[f] [⊠x]** = **260 cm²**. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[STACK-HACK]** Add 5 Faces via **[⊠+]**): Bottom (80) + 2 F/B (100) + 2 L/R (80) -> **[f] [⊠x]** = **260 cm²**. Way B (**[SPARK]** Full 6-Face Box Minus Missing Top): 340 - 80 = **260 cm²**!	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[STACK-HACK]** Add 5 Faces via **[⊠+]**): Bottom (80) + 2 F/B (100) + 2 L/R (80) -> **[f] [⊠x]** = **260 cm²**. Way B (**[SPARK]** Full 6-Face Box Minus Missing Top): 340 - 80 = **260 cm²**!

Day 35 (Lesson 8.2): Surface Area of Triangular Prisms & Pyramids ([f] [RECT] + [g] [TRI] + [□+])

SPIRAL PATH: Thread D (Inside-Out Chunking) • [[BOOMERANG] VISIT 5: CROSS-CHAPTER BOOMERANG — Revisits Day 21 TRI & Day 34 Nets] // VARIETY MODE: [SPARK] Mode C: Pyramid Tent Builder ({{key:TRI}} + {{key:□+}})

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): A square pyramid has 1 square floor (**4 by 4 = 16**) and 4 triangle roofs (**each area 6 -> 24**). Total is 16 + 24 = **40**! (**[ ] +5 Spark XP!**)

The Real Question: How do you keep a triangular prism (2 triangles + 3 rectangles) separated from a square pyramid (1 square base + 4 triangles) without mixing up their formulas?

Why Should I Care? (2-Min Hook): You don't need to memorize scary surface area formulas for prisms and pyramids! Just count the flat shapes on the net and feed them into **[□+]** using **[f] [RECT]** (**[f][7]**) for rectangular faces and **[g] [TRI]** (**[g][7]**) for triangular faces!
Why Try a New Way Today? (When the Old Way Gets Annoying): Pyramids and triangular prisms mix rectangles and triangles. Instead of tangling two formulas together, use **[f] [RECT]** for flat walls and **[g] [TRI]** for sloped roofs, tossing each into **[□+]** (**[STACK-HACK]**)!

3D Solid	What Its Unfolded Net Is Made Of	SC-6 [□+] Recipe
Square Pyramid (Camping Tent)	1 Square Base + 4 Identical Side Triangles	1. Base: **s [x²] [□+]** 2. 4 Triangles: **b [ENTER] h [g] [TRI] 4 [×] [□+]** 3. Total: **[f] [□x]**!
Triangular Prism (Toblerone Box)	2 Triangular Ends + 3 Rectangular Sides	1. 2 Ends: **b [ENTER] h [g] [TRI] 2 [×] [□+]** 2. 3 Sides: **L [ENTER] W [f] [RECT] [□+]** (for each side) 3. Total: **[f] [□x]**!

Problem 1 (Square Pyramid Tent): A square pyramid has a 10 m by 10 m square base and 4 triangular faces each with base 10 m and slant height 12 m. Find the total surface area.

1. [BLOCKS] TOUCH IT: **[ ]** 1 Square (10^2 = 100) + 4 Triangles (4 × 1/2 × 10 × 12 = 240).	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: **[g] [CL□]** -> **10 [x²] [□+]** (**100**) -> **10 [ENTER] 12 [g] [TRI] 4 [×] [□+]** (**240**) -> **[f] [□x]** = **340 m²**.	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

Problem 2 (Triangular Prism Ramp): A ramp has 2 right-triangle ends (base 6 ft, height 8 ft) and 3 rectangular faces that are 10 ft long with widths of 6 ft, 8 ft, and 10 ft. Find the total surface area.

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: **[g] [CL□]** -> 2 Triangles: **6 [ENTER] 8 [g] [TRI] 2 [×] [□+]** (**48**) -> Rect 1: **10 [ENTER] 6 [f] [RECT] [□+]** (**60**) -> Rect 2: **10 [ENTER] 8 [f] [RECT] [□+]** (**80**) -> Rect 3: **10 [ENTER] 10 [f] [RECT] [□+]** (**100**) -> **[f] [□x]** = **288 ft²**.	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

Problem 3 (Boss Level — Triangular Pyramid / Tetrahedron): A regular tetrahedron is made of 4 identical equilateral triangles, each with base 8 cm and height 6.93 cm. Find its total surface area!

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**8 [ENTER] 6.93 [g] [TRI] 4 [×]**): Snapshot **Y (1 Triangle): 27.72 | X: 4** -> **110.88 cm²**. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[STACK-HACK]** One Triangle × 4): **6 [ENTER] 5.2 [g] [TRI]** (**15.6**) **4 [×]** = **62.4 cm²**. Way B (**[SPARK]** Repeat **[□+]** 4 Times): Add **15.6** four times into **[□+]** -> **[f] [□x]** = **62.4 cm²**!	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[STACK-HACK]** One Triangle × 4): **6 [ENTER] 5.2 [g] [TRI]** (**15.6**) **4 [×]** = **62.4 cm²**. Way B (**[SPARK]** Repeat **[□+]** 4 Times): Add **15.6** four times into **[□+]** -> **[f] [□x]** = **62.4 cm²**!

Day 36 (Lesson 8.3): Volume of Rectangular Prisms with Whole & Fractional Edges ([f] [PRISM])

SPIRAL PATH: Thread D & B Boomerang • [[STACK-HACK] VISIT 3: STACK HACKER WAY — Revisits Day 2 Cubes, Day 7 Fractions & Day 20 Area] // VARIETY MODE: [BOOMERANG] Mode D: 1-Click 3D Volume ({{key:PRISM}})

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Build a 3 by 2 layer of cubes (6 cubes) and stack 4 layers high (24 cubes). Now type **3 [ENTER] 2 [ENTER] 4 [f] [PRISM]** (**[f][2]**) -> **24** in 1 click! (**[ ] +5 Spark XP!**)

The Real Question: Why do V = l × w × h and V = (Base Area B) × h give the exact same volume—and how does the SC-6 **[f] [PRISM]** key (**[f][9]**) compute volume from 3 stack floors at once?

Why Should I Care? (2-Min Hook): Look at key **9** on your SC-6: its yellow shift is **[f] [PRISM]** (**[f][9]**)! When you push Length (l), Width (w), and Height (h) onto Floors **Z**, **Y**, and **X** of the elevator (**1 [ENTER] w [ENTER] h**) and tap **[f] [PRISM]**, the SC-6 multiplies all three dimensions in a single click!
Why Try a New Way Today? (When the Old Way Gets Annoying): Packing physical unit cubes (**[BUILDER]**) shows why Volume is 3D space (l × w × h), while **[f] [PRISM]** (**[STACK-HACK]**) computes volume even when the edge lengths are fractions or decimals!

Method	How It Works	SC-6 Keystrokes	Result
Way 1: 3-Dimension **[f] [PRISM]** Key	Load **Z = 1**, **Y = w**, **X = h** and press **[f] [PRISM]** (**[f][9]**).	**4 [.] 1 [.] 2 [ENTER] 3 [ENTER] 2 [f] [PRISM]**	**27 in³**
Way 2: Layer Cake (B × h)	Find Base Floor Area B = l × w (**13.5 in²**), then stack h = 2 layers high!	**4.5 [ENTER] 3 [f] [RECT] 2 [×]**	**27 in³**
Way 3: Packing 1/2-inch Mini-Cubes	Each 1 in^3 holds 2^3 = 8 mini half-inch cubes. 27 × 8 = 216 mini-cubes!	**27 [ENTER] 2 [x²] [×]** -> **216** mini-cubes	**27 in³**

Problem 1 (Whole-Number Shipping Box): Find the volume of a crate measuring 12 in by 8 in by 5 in using [f] [PRISM].

1. [BLOCKS] TOUCH IT: **[ ]** Base layer of 12 × 8 = 96 cubes stacked 5 layers high.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**12 [ENTER] 8 [ENTER] 5 [f] [PRISM]**): Snapshot before **[f] [PRISM]**: **Z: 12 | Y: 8 | X: 5** -> **480 in³**.	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

Problem 2 (Fractional Edge Aquarium): A fish tank measures 31/2 ft long, 11/2 ft wide, and 2 ft tall. Find its volume in cubic feet!

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**3 [.] 1 [.] 2 [ENTER] 1 [.] 1 [.] 2 [ENTER] 2 [f] [PRISM]**): Snapshot **Z: 3 1/2 | Y: 1 1/2 | X: 2** -> **10 1/2 ft³** (**10.5 ft³**).	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

Problem 3 (Boss Level — Missing Water Depth): A rectangular planter has a base area of B = 45 in^2 (9 in × 5 in) and holds 360 in^3 of soil. How deep (h) is the soil?

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**360 [ENTER] 9 [ENTER] 5 [f] [RECT] [÷]**): Snapshot **Y (Volume): 360 | X (Base Area): 45** -> **8 inches deep**. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[STACK-HACK]** Base-Area Division): Since V = B × h, key **270 [ENTER] 45 [÷] = 6 in**. Way B (**[SPARK]** Forward **[f] [PRISM]** Proof with 9 × 5 × 6): **9 [ENTER] 5 [ENTER] 6 [f] [PRISM]** = **270 in³**!	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[STACK-HACK]** Base-Area Division): Since V = B × h, key **270 [ENTER] 45 [÷] = 6 in**. Way B (**[SPARK]** Forward **[f] [PRISM]** Proof with 9 × 5 × 6): **9 [ENTER] 5 [ENTER] 6 [f] [PRISM]** = **270 in³**!

Day 37 (Lesson 8.4): Circles, Cones & Right-Triangle Hypotenuse ([π], [f] [CIRC], [g] [CONE], [g] [HYP])

SPIRAL PATH: Thread C & D Boomerang • [[STACK-HACK] VISIT 4: CROSS-CHAPTER BOOMERANG — Revisits Day 2 Squares & Day 36 Volume] // VARIETY MODE: [DETECTIVE] Mode A: 3-4-5 Right Triangle Magic ({{key:HYP}} & {{key:CONE}})

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Lay a 3-tile row and a 4-tile column at a right angle. Measure the diagonal: it's 5 tiles (3^2+4^2=5^2)! Now press **3 [ENTER] 4 [g] [HYP]** (**[g][3]**) -> **5**! (**[ ] +5 Spark XP!**)

The Real Question: What are those cool geometry keys on **8** (**[f] [CIRC]**, **[g] [HYP]**) and **9** (**[g] [CONE]**)—and how do they give you superpowers in 7th and 8th grade geometry?

Why Should I Care? (2-Min Hook): Most middle schoolers dread circles (π r^2), cones (1/3π r^2 h), and the Pythagorean Theorem (√(a^2 + b^2)) because of all the squaring and π decimals. Your SC-6 has built-in 1-click geometry solvers right on keys **8** and **9**!
Why Try a New Way Today? (When the Old Way Gets Annoying): Typing **3.14** by hand causes rounding drift, and cones/right triangles take 4 steps on a basic calculator. Today you compare manual formula steps (**[SPARK]**) against the SC-6's 1-click **[f] [CIRC]**, **[g] [CONE]**, and **[g] [HYP]** keys (**[STACK-HACK]**)!

Geometry Goal	Formula	1-Click SC-6 Shortcut	Step-by-Step Stack Equivalent
Circle Area (r = 5)	A = π r^2	**5 [f] [CIRC]** (**[f][8]**) -> **78.54**	**5 [x²] [π] [×]** -> **78.54**
Right Triangle Hypotenuse (a=3, b=4)	c = √(a^2+b^2)	**3 [ENTER] 4 [g] [HYP]** (**[g][8]**) -> **5.00**	**3 [x²] 4 [x²] [+] [f] [√x]** -> **5**
Cone Volume (r = 3, h = 10)	V = 1/3π r^2 h	**3 [ENTER] 10 [g] [CONE]** (**[g][9]**) -> **94.25**	**3 [f] [CIRC] 10 [×] 3 [÷]** -> **94.25**

Problem 1 (Pizza Area Showdown): What is the area of a pizza with radius r = 6 in? Compute it with [f] [CIRC] AND check with 6 [x²] [π] [×].

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**6 [f] [CIRC]**): Snapshot **X: 113.10 in²** (36π ≈ 113.10).	3. [PROVE] VERIFIED SOLUTION: Divides by π (-> 36) and takes square root to restore radius **6**!

Problem 2 (Skate Ramp Diagonal [g] [HYP]): A right-triangle ramp is 12 ft long on the ground and 5 ft tall at the back. How long is the diagonal riding surface?

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**12 [ENTER] 5 [g] [HYP]**): Snapshot **Y: 12 | X: 5** -> **13 ft**.	3. [PROVE] VERIFIED SOLUTION: 144 + 25 = 169 -> √(169) = 13!

Problem 3 (Boss Level — Waffle Cone Volume): An ice cream cone has radius r = 3 cm and height h = 12 cm. Find its volume using [g] [CONE]!

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**3 [ENTER] 12 [g] [CONE]**): Snapshot **Y (Radius): 3 | X (Height): 12** -> **113.10 cm³**. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[STACK-HACK]** 1-Click **[g] [CONE]**): **3 [ENTER] 10 [g] [CONE]** = **94.25 cm³**. Way B (**[SPARK]** Cylinder ÷ 3 Formula): **[π] 3 [x²] [×] 10 [×] 3 [÷]** = **94.25 cm³**!	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[STACK-HACK]** 1-Click **[g] [CONE]**): **3 [ENTER] 10 [g] [CONE]** = **94.25 cm³**. Way B (**[SPARK]** Cylinder ÷ 3 Formula): **[π] 3 [x²] [×] 10 [×] 3 [÷]** = **94.25 cm³**!

CHAPTER 9: Statistical Measures & Data Displays (Days 38–41)

Aligned to Big Ideas Math Grade 6, Chapters 9 & 10: Mean, Median, Mode, Range, Quartiles, Interquartile Range (IQR), Box-and-Whisker Plots, and Mean Absolute Deviation (MAD).

Day 38 (Lesson 9.1): Dot Plots, Frequency & Mean as "Fair Share Leveling" ([□+], [g] [x̄], [FRQ])

SPIRAL PATH: Thread F (Balance & Fair Share) • [[SPARK] VISIT 2: PATTERN WAY — Revisits Day 1 Division & Day 22 □+] // VARIETY MODE: [SPARK] Mode C: Level the Cube Towers ({{key:□+}} & {{key:x̄}})

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Build 3 cube towers of heights **2**, **4**, and **6**. Move 2 cubes from the **6**-tower to the **2**-tower—now all three towers are height **4**! Press **[g] [CL□] 2 [□+] 4 [□+] 6 [□+] [g] [x̄]** -> **4**! (**[ ] +5 Spark XP!**)

The Real Question: What does the "Mean" (average) physically look like with blocks—and how do you enter a whole data set into **[□+]** without losing your place?

Why Should I Care? (2-Min Hook): On paper, finding the Mean of 8 numbers means adding them all up, hoping you didn't skip one, and dividing by 8. On the SC-6:
Why Try a New Way Today? (When the Old Way Gets Annoying): What is an average (Mean)? If you build unequal towers of cubes (**[BUILDER]**) and move cubes from tall towers to short ones until every tower is the same height, that leveled height is the Mean! And for big lists, **[□+]** + **[g] [x̄]** (**[STACK-HACK]**) levels them in 1 click!

Camera Angle	What You Do	What You See
Angle 1: Physical "Fair Share" Leveling (**COUNT-01**)	Build 5 towers of heights **3, 7, 5, 9, 6**. Move tiles from the tallest towers (**9** and **7**) onto the shortest towers (**3** and **5**) until all 5 towers are the exact same height!	Every tower stands 6 tiles tall! That leveled-out height IS the Mean (**x̄ = 6**)!
Angle 2: Textbook Formula (Σ x/n)	Add 3+7+5+9+6 = 30, then divide by 5 towers: 30 ÷ 5 = 6.	Total sum **30** (**[f] [□x]**) divided by count **5** (**[N]**) = 6.
Angle 3: SC-6 1-Key **[g] [x̄]**	**[g] [CL□]** -> **3 [□+] 7 [□+] 5 [□+] 9 [□+] 6 [□+]** -> press **[g] [x̄]** (**[g][N]**).	Screen shows **6.00** immediately!

Problem 1 (Gaming Score Average): Your 5 match scores are 18, 24, 15, 27, 21. Load them with [□+] and record [N], [f] [□x], and [g] [x̄].

1. [BLOCKS] TOUCH IT: **[ ]** Leveled 5 scores to their fair-share average.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: **[g] [CL□]** -> enter 5 scores with **[□+]**. Snapshot: **N: 5 | □x: 105 | x̄: 21**.	3. [PROVE] VERIFIED SOLUTION: 21 × 5 = 105!

Problem 2 (The Typo Rescue with [f] [□-]): Suppose you entered 10, 20, 35 with [□+], and then realized 35 was supposed to be 25! Fix it without clearing the whole list!

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: Type **35 [f] [□-]** (**[f][□+]**, removes **35** so **n = 2**), then type **25 [□+]** (**n = 3**). Press **[g] [x̄]** -> **18.33** (55/3).	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

Problem 3 (Boss Level — Missing Test Score for an A): You need a Mean of 90 across 4 quizzes (4 × 90 = 360 total points). Your first 3 quizzes are 86, 92, and 88. What score do you need on Quiz 4?

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: **[g] [CL□]** -> **86 [□+] 92 [□+] 88 [□+]** -> **90 [ENTER] 4 [×] [f] [□x] [-]** (**360 - 266**) -> **94**. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[STACK-HACK]** Target Total Minus **[f] [□x]**): Need 4 × 90 = 360 pts; current **[f] [□x]** = 265 -> 360 - 265 = 95. Way B (**[BUILDER]** Fair-Share Points Above/Below 90): (-6) + (-2) + (+3) = -5 deficit, so Quiz 4 needs 90 + 5 = 95!	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[STACK-HACK]** Target Total Minus **[f] [□x]**): Need 4 × 90 = 360 pts; current **[f] [□x]** = 265 -> 360 - 265 = 95. Way B (**[BUILDER]** Fair-Share Points Above/Below 90): (-6) + (-2) + (+3) = -5 deficit, so Quiz 4 needs 90 + 5 = 95!

Day 39 (Lesson 9.2): Median, Mode & Range — One Data Entry, Five Instant Answers!

SPIRAL PATH: Thread F (Balance & Fair Share) • [[STACK-HACK] VISIT 3: STACK HACKER WAY — Revisits Day 38 Mean] // VARIETY MODE: DICE Mode E: Live Dice Roll Stats Lab ({{key:MED}}, {{key:MODE}}, {{key:RNG}})

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): With **2, 4, 6** still in memory from yesterday, press **[f] [MED]** (**[g][7]**) -> **4** (middle number) and **[RNG]** (**[g][8]**) -> **4** (6 - 2)! Zero re-typing! (**[ ] +5 Spark XP!**)

The Real Question: Why does the textbook make you sort all the numbers from least to greatest just to find the Median, Mode, Min, Max, and Range—and how does the SC-6 give you all 5 from the exact same **[Σ+]** list?

Why Should I Care? (2-Min Hook): Here is the best part of Row 2 on the SC-6: once you enter your data list with **[Σ+]** in ANY order (you don't even have to sort them by hand first!), the calculator sorts them inside its chip and unlocks every single measure of center and spread with one tap:
Why Try a New Way Today? (When the Old Way Gets Annoying): On Day 38 we leveled towers to find the Mean (**[g] [x̄]**). Why do we also need the Median (**[f] [MED]**)? Because one crazy outlier (like a $450 console in a list of $20 games) drags the Mean way up, while the middle number (**[f] [MED]**) stays rock-solid!

Problem 1 (1-Entry, 5-Answer Sweep): Enter the shoe sizes 7, 9, 6, 7, 10, 8, 9, 7 using [g] [CLΣ] and [Σ+]. Read off all 5 stats!

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: **Mean ([g] [x̄])**: **7.88** | **Median ([f] [MED])**: **7.5** | **Mode ([g] [MODE])**: **7** | **Min ([f] [MIN])**: **6** | **Max ([g] [MAX])**: **10** | **Range ([RNG])**: **4**.	3. [PROVE] VERIFIED SOLUTION: Sorted list is **6, 7, 7, 7, 8, 9, 9, 10**—middle two are **7** and **8** (average **7.5**), and 10 - 6 = 4!

Problem 2 (Outlier Detective): Daily dog-walking earnings for a week are 15, 18, 15, 20, 17, 90. Find both the Mean and the Median. Which one better represents a typical day?

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: **Mean ([g] [x̄])**: **29.17** | **Median ([f] [MED])**: **17.50**.	3. [PROVE] VERIFIED SOLUTION: **Median (17.50)** is much better because the **90** outlier makes **$29.17** higher than 5 out of the 6 days!

Problem 3 (Boss Level — Dice Roll Experiment): Roll your physical Probability Die (PROB-01) (or press [DICE]) 7 times, log each roll with [Σ+], and write your [f] [MED], [g] [MODE], and [RNG]!

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[BUILDER]** Line Up Physical Tiles in Order): Sort your 5 rolls from least to greatest to spot the middle tile (**Median**) and **Max - Min** (**Range**). Way B (**[STACK-HACK]** SC-6 **[f] [MED]** & **[RNG]** Keys): Load into **[Σ+]** and press **[f] [MED]** and **[RNG]**—100% match!	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[BUILDER]** Line Up Physical Tiles in Order): Sort your 5 rolls from least to greatest to spot the middle tile (**Median**) and **Max - Min** (**Range**). Way B (**[STACK-HACK]** SC-6 **[f] [MED]** & **[RNG]** Keys): Load into **[Σ+]** and press **[f] [MED]** and **[RNG]**—100% match!

Day 40 (Lesson 9.3): Quartiles, Interquartile Range ([IQR]) & Box-and-Whisker Plots in 5 Seconds

SPIRAL PATH: Thread F (Balance & Fair Share) • [[STACK-HACK] VISIT 4: BOSS REVISIT — Revisits Day 6 Quarters & Day 39 Median] // VARIETY MODE: [BOOMERANG] Mode D: 3-Second Box-Plot Builder ({{key:1Q}}, {{key:3Q}}, {{key:IQR}})

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Load **10, 20, 30, 40, 50** into **[Σ+]**. Tap **[f] [MIN]** (**10**), **[f] [1Q]** (**15**), **[f] [MED]** (**30**), **[g] [3Q]** (**45**), **[g] [MAX]** (**50**), and **[IQR]** (**30**)! Whole Box Plot in 5 seconds! (**[ ] +5 Spark XP!**)

The Real Question: How do you draw a Box-and-Whisker Plot (5-Number Summary) without getting confused about Q1, Median, and Q3?

Why Should I Care? (2-Min Hook): A Box-and-Whisker Plot only needs 5 numbers (called the 5-Number Summary):
Why Try a New Way Today? (When the Old Way Gets Annoying): Finding Q_1, Q_3, and IQR by hand requires splitting a sorted list in half and finding medians of each half. Once your data is in **[Σ+]** from Day 39, pressing **[f] [1Q]**, **[g] [3Q]**, and **[IQR]** (**[STACK-HACK]**) builds your entire Box Plot in 3 seconds!

Problem 1 (Instant 5-Number Summary): Clear stats ([g] [CLΣ]) and enter the 9 test scores: 68, 72, 75, 78, 82, 85, 88, 92, 96. Record the 5-Number Summary and [IQR].

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: **MIN**: **68** | **1Q**: **73.5** | **MED**: **82** | **3Q**: **90** | **MAX**: **96** | **[IQR]**: **16.5**.	3. [PROVE] VERIFIED SOLUTION: Q_3 - Q_1 = 90 - 73.5 = 16.5!

Problem 2 (What Percent Lives in the Box?): In any Box-and-Whisker Plot, each of the 4 sections (Left Whisker, Left Half of Box, Right Half of Box, Right Whisker) holds 25\% (1/4) of the data!

1. [BLOCKS] TOUCH IT: **[ ]** Four 1/4 Fraction Studio blocks make 1 Whole (100\%).	2. [STACK] KEYSTROKES & Y|X: Verified on 4-line XYZT stack.	3. [PROVE] VERIFIED SOLUTION: The middle box (from **1Q** to **3Q**) holds 25\% + 25\% = 50\% of the data!

Problem 3 (Boss Level — Comparing Consistency with [IQR]): Player A has an IQR of 4 points and Player B has an IQR of 19 points. Which player is more consistent (less spread out)?

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[BUILDER]** Box-Plot Width Visual): A narrower middle box (**IQR = 4** vs **18**) means the middle 50\% of scores are tightly clustered (Player A). Way B (**[STACK-HACK] 3Q - 1Q** Stack Check): Verify **[g] [3Q]** minus **[f] [1Q]** equals **[IQR]**!	3. [PROVE] VERIFIED SOLUTION: Player A (smaller **IQR** = tighter box = much more consistent!).

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[BUILDER]** Box-Plot Width Visual): A narrower middle box (**IQR = 4** vs **18**) means the middle 50\% of scores are tightly clustered (Player A). Way B (**[STACK-HACK] 3Q - 1Q** Stack Check): Verify **[g] [3Q]** minus **[f] [1Q]** equals **[IQR]**!

Day 41 (Lesson 9.4): Mean Absolute Deviation ([g] [MAD]) Without the 20-Minute Subtraction Nightmare

SPIRAL PATH: Thread E & F Boomerang • [[BOOMERANG] VISIT 5: CROSS-CHAPTER BOOMERANG — Revisits Day 26 |x| & Day 38 Mean] // VARIETY MODE: [BOOMERANG] Mode D: Beat the 20-Minute Homework Monster ({{key:MAD}})

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Towers **2, 4, 6** have Mean **4**. Their distances from **4** are **2, 0, 2** (average distance = 4 ÷ 3 = 1.33). Now press **[g] [MAD]** (**[g][5]**) on your SC-6 -> **1.33** in 1 click! (**[ ] +5 Spark XP!**)

The Real Question: Why is MAD the most dreaded homework problem in 6th grade—and how does the SC-6 **[g] [MAD]** key (**[g][⊠+]**) turn 20 minutes of tedious subtraction into a 3-second check?

Why Should I Care? (2-Min Hook): Be honest: every 6th grader on Earth hates Mean Absolute Deviation (MAD). Why? Because to find the MAD of 6 numbers by hand, the textbook makes you:
Why Try a New Way Today? (When the Old Way Gets Annoying): Mean Absolute Deviation (**MAD**) is the most dreaded calculation in 6th grade because doing it by hand takes 15 steps (find mean, subtract every number, take **[|x|]** from Day 26, and average again). Today you see how Day 26's **[|x|]** + Day 38's **[g] [x̄]** build **MAD**—and unlock the 1-click **[g] [MAD]** key (**[STACK-HACK]**)!

Problem 1 (Instant MAD Check): Find the Mean ([g] [x̄]) and Mean Absolute Deviation ([g] [MAD]) of 10, 12, 14, 16, 18.

1. [BLOCKS] TOUCH IT: **[ ]** Mean is **14**. Distances from 14 are **4, 2, 0, 2, 4**.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: **[g] [CL⊠]** -> enter 5 numbers with **[⊠+]**. **[g] [x̄]** = **14.00** | **[g] [MAD]** = **2.40** (4+2+0+2+4/5 = 12/5 = 2.4).	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

Problem 2 (Real-Life Decimal MAD): Find the Mean and MAD of the 6 sprint times (in seconds): 12.4, 13.0, 12.6, 13.2, 12.8, 14.0.

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: **[g] [CL⊠]** -> enter all 6 times with **[⊠+]**. Snapshot: **[g] [x̄]**: **13.00 s** | **[g] [MAD]**: **0.40 s**.	3. [PROVE] VERIFIED SOLUTION: Distances from **13.0** are **0.6, 0.0, 0.4, 0.2, 0.2, 1.0** (Sum = 2.4 ÷ 6 = 0.40)!

Problem 3 (Boss Level — Full Stats Sweep): For the data 5, 8, 8, 10, 14, record [g] [x̄], [f] [MED], [g] [MODE], [RNG], [IQR], and [g] [MAD] from a single [⊠+] entry!

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: **x̄**: **9.00** | **MED**: **8.00** | **MODE**: **8.00** | **RNG**: **9.00** | **IQR**: **5.50** (**1Q=6.5, 3Q=12**) | **MAD**: **2.40**. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[STACK-HACK]** 1-Click **[g] [MAD]** Key): Load **5, 8, 8, 10, 14** into **[⊠+]** and press **[g] [MAD]** = **2.4**. Way B (**[SPARK]** Distance-from-Mean Check using Day 26 **[|x|]**): Mean = 9; distances 4+1+1+1+5 = 12 -> 12 ÷ 5 = **2.4**!	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[STACK-HACK]** 1-Click **[g] [MAD]** Key): Load **5, 8, 8, 10, 14** into **[⊠+]** and press **[g] [MAD]** = **2.4**. Way B (**[SPARK]** Distance-from-Mean Check using Day 26 **[|x|]**): Mean = 9; distances 4+1+1+1+5 = 12 -> 12 ÷ 5 = **2.4**!

CHAPTER 10: Grade 7 Bridge — Signed Rationals, 2-Step Equations & Probability (Days 42–45)

Aligned to Big Ideas Math Grade 7, Chapters 1–3 & Chapter 7: Adding, Subtracting, Multiplying & Dividing Signed Rational Numbers, Solving Two-Step Equations, and Probability/Combinatorics.

Day 42 (Lesson 10.1): Adding & Subtracting Signed Numbers — Zero Pairs & Why "Minus a Negative" Adds!

SPIRAL PATH: Thread E (Zero-Pair & Distance) • [[BOOMERANG] VISIT 5: GRADE 7 BRIDGE BOOMERANG — Revisits Days 7, 24 & 26] // VARIETY MODE: [DETECTIVE] Mode A: Penalty Cancellation Trick (10 - (-4) = 14)

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Your game score is **10**, and the ref erases a -4 penalty (10 - (-4)). Erasing a penalty makes your score go UP to **14**! Key **10 [ENTER] 4 [+/-] [-]** -> **14**! (**[ ] +5 Spark XP!**)

The Real Question: Why does -8 + 5 = -3, and why does subtracting a negative like 10 - (-4) turn into +14?

Why Should I Care? (2-Min Hook): In 7th grade, almost every math error comes from negative sign rules. With your Two-Color Counters (**BASE-10**) (Yellow = +1, Red = -1):
Why Try a New Way Today? (When the Old Way Gets Annoying): In Chapter 6 (**Day 24**), you met Yellow (+1) and Red (-1) counters. Today they boomerang back for 7th-grade signed arithmetic: removing red debt counters (**[BUILDER]**) proves why subtracting a negative (10 - (-4)) is the exact same as adding a positive (10 + 4 = 14)!

Problem	Physical Bicolor Counter View (BASE-10)	SC-6 Keystrokes	Result & [f] [LASTX] Proof
-8 + 5	8 Red (-) + 5 Yellow (+) -> 5 Zero Pairs cancel, leaving 3 Red!	**8 [+/-] [ENTER] 5 [+]**	**-3** (**[f] [LASTX] [-]** -> -8)
-6 + (-4)	6 Red (-) + 4 more Red (-) -> 10 Red (Zero cancellation!).	**6 [+/-] [ENTER] 4 [+/-] [+]**	**-10** (**[f] [LASTX] [-]** -> -6)
10 - (-4)	Remove 4 Red debt tokens = add 4 Yellow (10 + 4)!	**10 [ENTER] 4 [+/-] [-]**	**14** (**[f] [LASTX] [+]** -> 10)

Problem 1 (Zero-Pair Battle): Solve -15 + 9 and -15 + (-9) side by side so you see the difference!

1. [BLOCKS] TOUCH IT: **[ ]** Opposite signs fight (cancel 9 pairs -> -6); same signs team up (-> -24).	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: **15 [+/-] [ENTER] 9 [+]** -> **-6**. **15 [+/-] [ENTER] 9 [+/-] [+]** -> **-24**.	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

Problem 2 (Taking Away a Penalty): Solve 12 - (-8) and -12 - (-8) on the SC-6.

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: **12 [ENTER] 8 [+/-] [-]** -> **20** (12 + 8). **12 [+/-] [ENTER] 8 [+/-] [-]** -> **-4** (-12 + 8).	3. [PROVE] VERIFIED SOLUTION: -4 + (-8) = -12!

Problem 3 (Boss Level — Signed Fractions): Solve -3/4 - (-5/8).

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**[.] 3 [.] 4 [+/-] [ENTER] [.] 5 [.] 8 [+/-] [-]**): Snapshot **Y: -3/4 | X: -5/8** -> **-1/8**. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[STACK-HACK]** Signed Fraction Stack): **[.] 3 [.] 4 [+/-] [ENTER] [.] 5 [.] 8 [+/-] [-]** = **-1/8**. Way B (**[SPARK]** Add-the-Opposite + Common 8ths): -6/8 + (+5/8) = **-1/8**!	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[STACK-HACK]** Signed Fraction Stack): **[.] 3 [.] 4 [+/-] [ENTER] [.] 5 [.] 8 [+/-] [-]** = **-1/8**. Way B (**[SPARK]** Add-the-Opposite + Common 8ths): -6/8 + (+5/8) = **-1/8**!

Day 43 (Lesson 10.2): Multiplying & Dividing Signed Rational Numbers — The Light-Switch Flip Rule

SPIRAL PATH: Thread E & B Boomerang • [[BOOMERANG] VISIT 6: GRADE 7 BRIDGE BOOMERANG — Revisits Days 8, 24 & 42] // VARIETY MODE: [SPARK] Mode C: Light-Switch Double-Flip ({{key:+/-}} × {{key:+/-}})

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Type **5** on your SC-6. Tap **[+/-]** twice: **5 -> -5 -> +5**! Two negative flips always bring you back to POSITIVE! Now key **3 [+/-] [ENTER] 4 [+/-] [×]** -> **+12**! (**[ ] +5 Spark XP!**)

The Real Question: Why does a Negative times a Negative equal a POSITIVE—and how do you tell the sign of any multiplication or division problem in 1 second?

Why Should I Care? (2-Min Hook): Think of a negative sign (**-**) as flipping a light switch (just like the **[+/-]** key!):
Why Try a New Way Today? (When the Old Way Gets Annoying): Why does (-)×(-) = (+)? Pressing **[+/-]** twice (**[BUILDER]**) flips a negative right back to positive! Combining Day 8's Reciprocal Flip (**[f] [1/x]**) with today's Even/Odd Sign-Flip rule (**[SPARK]**) lets you predict the sign of any product or quotient before touching a key!

Problem 1 (The 4 Sign Combos): Compute 6 × (-4), (-6) × 4, and (-6) × (-4) on the SC-6.

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: **6 [ENTER] 4 [+/-] [×]** -> **-24**. **6 [+/-] [ENTER] 4 [+/-] [×]** -> **+24**.	3. [PROVE] VERIFIED SOLUTION: 24 ÷ (-4) = -6!

Problem 2 (Negative Base Squared): Compute (-5)^2 on the SC-6 (5 [+/-] [x²]). Why is it +25 instead of -25?

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**5 [+/-] [x²]**): Result is **+25** because (-5) × (-5) has TWO negative flips!	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

Problem 3 (Boss Level — Signed Fraction Division): Solve -5/6 ÷ (-5/12).

1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**[.] 5 [.] 6 [+/-] [ENTER] [.] 5 [.] 12 [+/-] [÷]**): Negative divided by negative = Positive **+2**! • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[STACK-HACK]** Signed Fraction Stack): **[.] 5 [.] 6 [+/-] [ENTER] [.] 5 [.] 12 [+/-] [÷]** = **+2**. Way B (**[SPARK]** Sign-Flip Rule + Day 8 Reciprocal): 2 negatives = (+), and 5/6 × 12/5 = **+2**!	3. [PROVE] VERIFIED SOLUTION: 2 × (-5/12) = -10/12 = -5/6!

[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[STACK-HACK]** Signed Fraction Stack): **[.] 5 [.] 6 [+/-] [ENTER] [.] 5 [.] 12 [+/-] [÷]** = **+2**. Way B (**[SPARK]** Sign-Flip Rule + Day 8 Reciprocal): 2 negatives = (+), and 5/6 × 12/5 = **+2**!

Day 44 (Lesson 10.3): Solving Two-Step Equations (ax + b = c) — Unwrapping the Present in Reverse!

SPIRAL PATH: Thread F, B & D Boomerang • [[STACK-HACK] VISIT 6: GRADE 7 BOSS BOOMERANG — Revisits Days 3, 30, 31 & 42] // VARIETY MODE: [TARGET] Mode B: Unwrap the Present in Reverse (ax + b = c)

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): To Solve 2x + 3 = 11: first untie the loose ribbon (**11 [ENTER] 3 [-]** -> **8**), then open the 2 boxes (**2 [÷]** -> **x = 4**)! Two clicks, zero algebra rewrites! (**[ ] +5 Spark XP!**)

The Real Question: In a two-step equation like 3x + 7 = 28, how do you know whether to undo the +7 or the × 3 FIRST—and how do you show your work in a 2-number stack snapshot?

Why Should I Care? (2-Min Hook): Think about wrapping a birthday gift: first you put the toy in a box (× 3), and then you tie a ribbon (+ 7) around the outside. When you open the gift, what do you undo first? You untie the loose ribbon (**- 7**) FIRST, and open the box (**÷ 3**) SECOND!
Why Try a New Way Today? (When the Old Way Gets Annoying): Two-step equations (ax + b = c) combine Day 30 (Addition/Subtraction Undo), Day 31 (Multiplication/Division Undo), and Day 42 (Signed Numbers) into one reverse unwrap! Solve by sweeping blocks off the Algebra Stage (**[BUILDER]**) OR running the 2-step reverse stack (**[STACK-HACK]**)!

Step	Physical Algebra Stage (ALGB-01)	SC-6 Elevator Keystrokes	Stack Snapshot (Y | X)
Setup	3 green x-rods + 7 yellow unit cubes = 28 unit cubes.	**28 [ENTER]**	**Y: 28 | X: 28**
Step 1: Clear Loose Cubes	Take **7** unit cubes off both pans! Now 3 x-rods = **21** cubes.	**7 [-]**	**X: 21** (Step 1 work: 3x = 21)
Step 2: Split Across Rods	Divide **21** cubes equally among the **3** x-rods!	**3 [÷]**	**X: 7** (Solution: x = 7!)
Step 3: 3-Click Proof	Plug x = 7 back in: 3(7) + 7 = 28!	**3 [×] 7 [+]**	Returns to **28**!

Problem 1 (Clear Added Cubes First): Solve 4x + 9 = 45. Write the Step 1 snapshot (4x = ?) and the final answer!		
1. [BLOCKS] TOUCH IT: **[ ]** Removed 9 cubes from 45 (-> 36), then split across 4 x-rods.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**45 [ENTER] 9 [-] 4 [÷]**): After **[-]** (**4x = 36**), before **[÷]**: **Y: 36 | X: 4** -> **x = 9**.	3. [PROVE] VERIFIED SOLUTION: 9 × 4 + 9 = 45!

Problem 2 (Clear Subtracted Cubes First): Solve 5x - 12 = 38.		
1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**38 [ENTER] 12 [+] 5 [÷]**): Step 1 (**5x = 50**), Snapshot before **[÷]**: **Y: 50 | X: 5** -> **x = 10**.	3. [PROVE] VERIFIED SOLUTION: 10 × 5 - 12 = 38!

Problem 3 (Boss Level — Negative Coefficient Two-Step): Solve -3x + 14 = 2.		
1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**2 [ENTER] 14 [-] 3 [+/-] [÷]**): Step 1: 2 - 14 = -12 (**-3x = -12**). Snapshot before **[÷]**: **Y: -12 | X: -3** -> **x = 4**. • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (**[STACK-HACK]** 2-Step Reverse Stack): **2 [ENTER] 14 [-]** (**-12**) **3 [+/-] [÷]** = **x = 4**. Way B (**[BUILDER]** Plug-In Forward Check from Day 16): Substitute x=4 into -3(4) + 14 = -12 + 14 = **2**!	3. [PROVE] VERIFIED SOLUTION: -3(4) + 14 = -12 + 14 = 2!
[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (**[STACK-HACK]** 2-Step Reverse Stack): **2 [ENTER] 14 [-]** (**-12**) **3 [+/-] [÷]** = **x = 4**. Way B (**[BUILDER]** Plug-In Forward Check from Day 16): Substitute x=4 into -3(4) + 14 = -12 + 14 = **2**!		

Day 45 (Lesson 10.4): Probability, Combinatorics & The Graduation Boss Sprint ([DICE], [x!], [f] [nCr], [f] [?])

SPIRAL PATH: All 6 Threads Finale • [[BADGE] GRADUATION BOSS SPRINT — Revisits Fractions (Ch. 2), Percents (Ch. 3) & Stats (Ch. 9)] // VARIETY MODE: DICE Mode E: Graduation Live Dice & {{key:?}} Quiz Sprint

[SPARK] 30-SECOND ZERO-FAIL STARTER (Instant Win #1): Press **6 [f] [1…x]** (**[f][DICE]**) three times to roll a digital 6-sided die, then press **3 [x!]** (**[g][DICE]**) -> **6** ways to arrange 3 trophies! (**[ ] +5 Spark XP!**)

The Real Question: How do you calculate the odds of winning a game (**[DICE]**), count how many teams or passwords are possible (**[x!]**, **[f] [nCr]**, **[g] [nPr]**), and prove you've mastered middle school math on the SC-6 Quiz Engine (**[f] [?]**)?

Why Should I Care? (2-Min Hook): You made it to Day 45! Today you unlock the top-left superpower keys on the SC-6 (**[DICE]**, **[x!]**, **[f] [nCr]**, **[g] [nPr]**) and take the Graduation Boss Sprint on the built-in Math Quiz Engine (**[f] [?]**, **[f][C]**)!
Why Try a New Way Today? (When the Old Way Gets Annoying): On Graduation Day, you combine physical dice/spinner trials (**[BUILDER]**), 3-way fraction/decimal/percent probability (**[SPARK]**), and the SC-6's combinatorics (**[x!]**, **[f] [nCr]**, **[g] [nPr]**) and Quiz Engine (**[f] [?] [STACK-HACK]**)!

Problem 1 (3-Way Spinner Probability): A 10-sector spinner (PROB-01) has 4 Blue sectors, 3 Gold sectors, and 3 Red sectors. Find the probability of spinning Blue as a simplified fraction, decimal, and percent!		
1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (**[.] 4 [.] 10 [ENTER] [g] [SIMP]**): Fraction: **2/5**. Press **[FDISP]** -> Decimal: **0.40**. Press **[f] [%]** -> Percent: **40%**.	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

Problem 2 (Podium vs. Pizza Toppings): From 10 pizza toppings, how many ways can you choose 3 toppings where order does NOT matter ([f] [nCr])? Compare that to choosing 1st, 2nd, and 3rd place from 10 runners ([g] [nPr]).		
1. [BLOCKS] TOUCH IT: Hands-on manipulative build verified.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It: Toppings (**10 [ENTER] 3 [f] [nCr]**): **120 combinations**. Podium (**10 [ENTER] 3 [g] [nPr]**): **720 permutations**.	3. [PROVE] VERIFIED SOLUTION: Verified via [f][LASTX] inverse operation.

Problem 3 (GRADUATION BOSS SPRINT — SC-6 Quiz Engine `[f] [?]`): Launch the built-in SC-6 Math Quiz Engine (`[f] [?]`, `[f][C]`), complete a 10-question streak using your 4-line stack and physical blocks, and compute the number of ways to arrange 5 Boss Badges (`5 [x!]`).		
1. [BLOCKS] TOUCH IT: `[ ]` Arranged 5 physical kit badges in a row.	2. [STACK] KEYSTROKES & Y|X: [STACK] Stack It (`5 [x!]`): Snapshot `X: 120` ways (5 × 4 × 3 × 2 × 1 = 120). • [2x-STAR] Twin-Way Bonus (+25 XP toward [SKIP-SHIELD] Skip Shield): Way A (`[STACK-HACK]` Factorial Key `[x!]`): `5 [x!]` = 120 ways. Way B (`[SPARK]` Permutation Key `[g] [nPr]`): Choosing all 5 badges in order is `5 [ENTER] 5 [g] [nPr]` = 120 ways (5× 4× 3× 2× 1)!	3. [PROVE] VERIFIED SOLUTION: Completed 10-question SC-6 Quiz Engine Graduation Streak!
[2x-STAR] TWIN-WAY REVISIT KEY (Way A == Way B): Way A (`[STACK-HACK]` Factorial Key `[x!]`): `5 [x!]` = 120 ways. Way B (`[SPARK]` Permutation Key `[g] [nPr]`): Choosing all 5 badges in order is `5 [ENTER] 5 [g] [nPr]` = 120 ways (5× 4× 3× 2× 1)!		